Apalachicola Bay Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

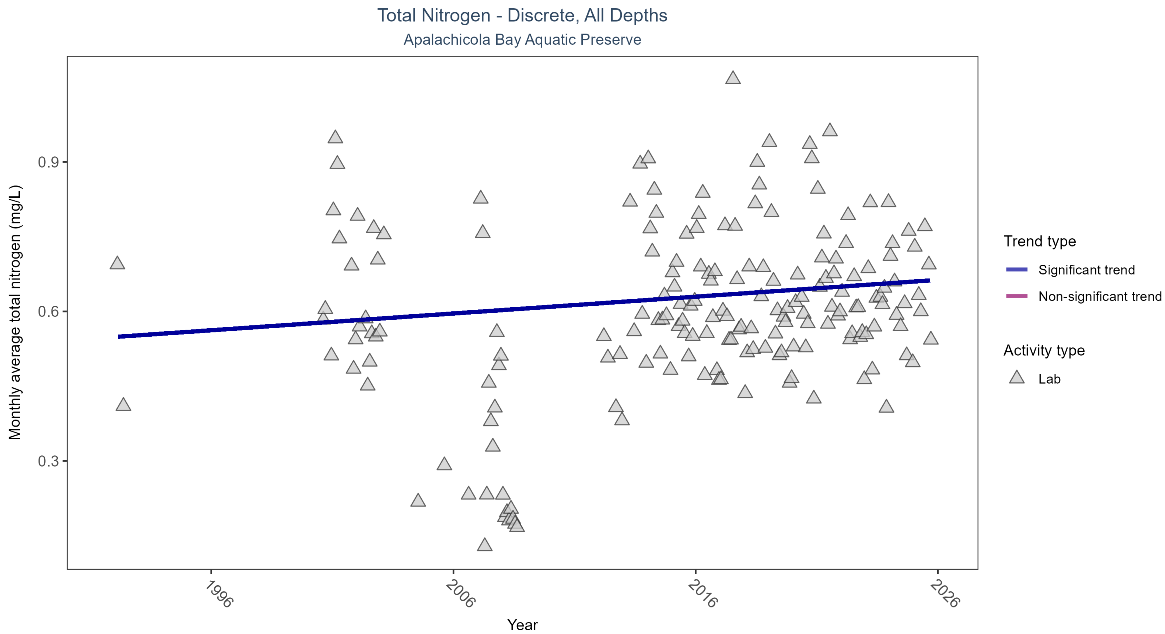

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2873	25	1992 - 2025	0.612	0.1175	0.54867	0.00337	0.0286

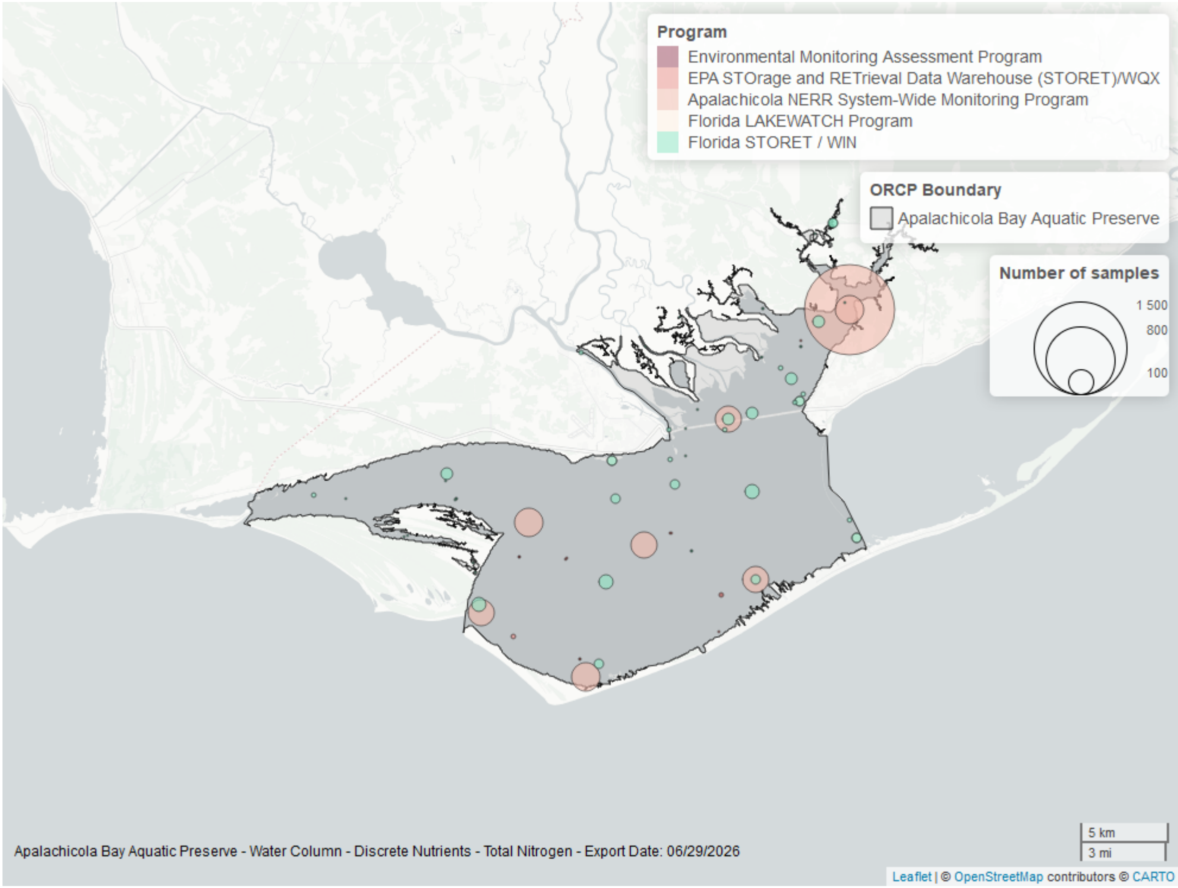

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

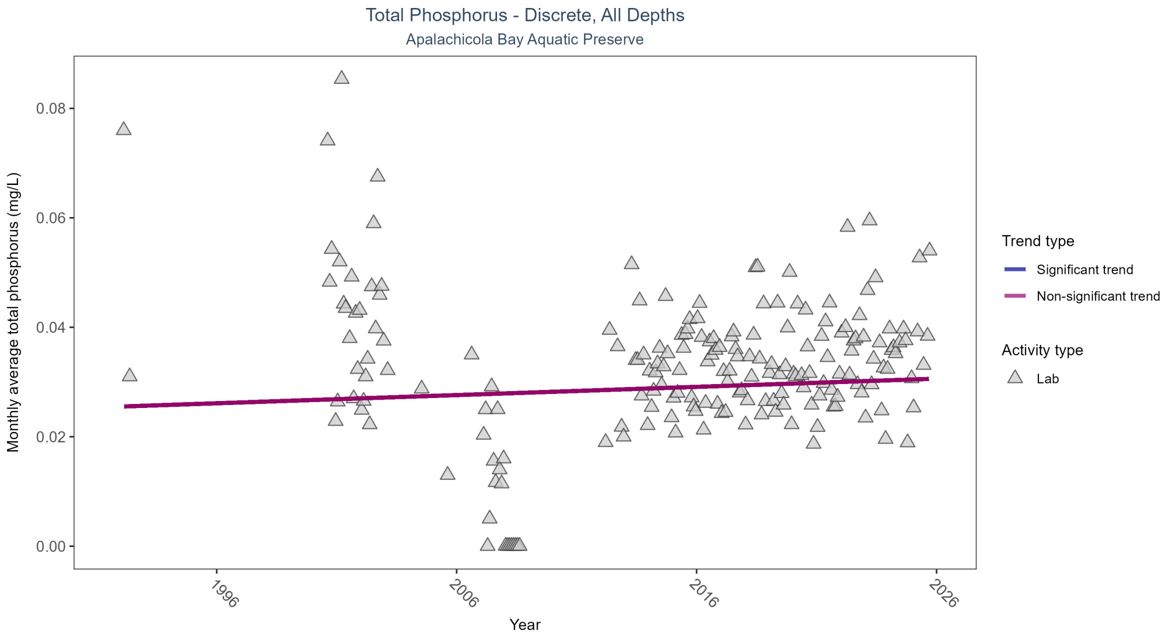

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1992 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2976	24	1992 - 2025	0.031	0.06685	0.02551	0.00015	0.2553

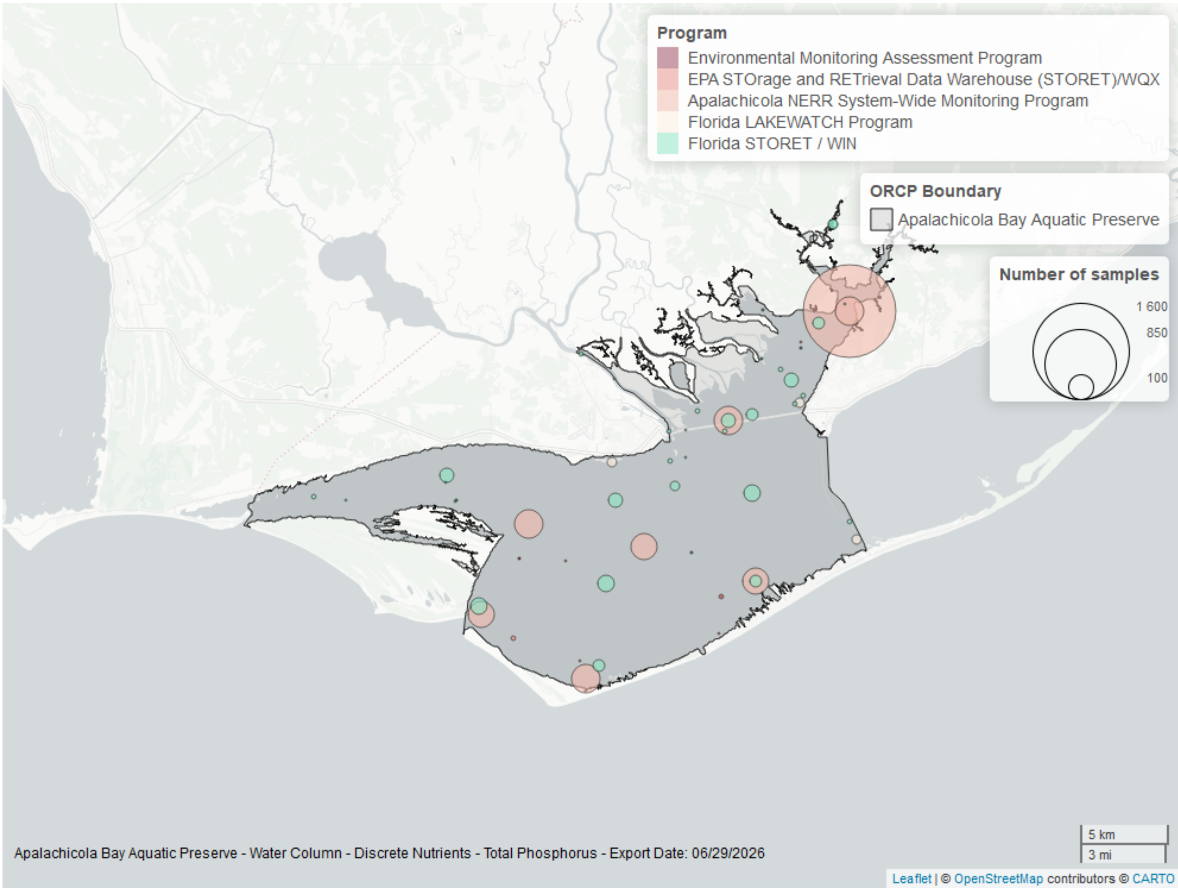

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

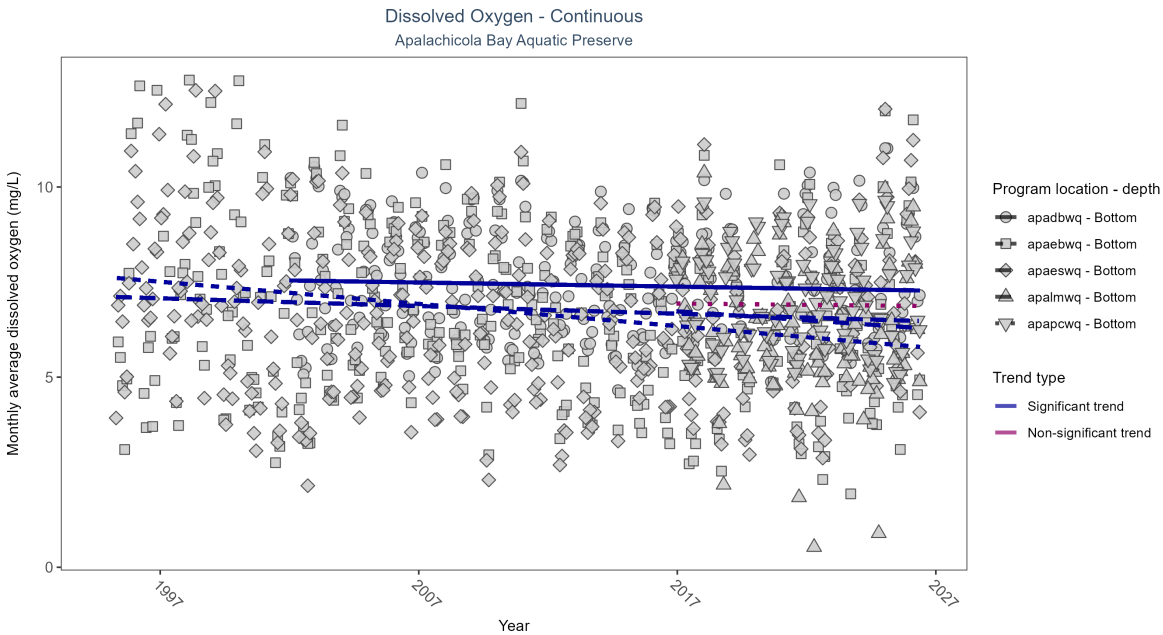

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At four program locations, monthly average dissolved oxygen decreased between 0.01 and 0.06 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Decreasing trend	661293	25	2002 - 2026	7.3	-0.10	7.54	-0.01	0.02
apaebwq	Decreasing trend	705983	32	1995 - 2026	6.8	-0.28	7.63	-0.06	0.00
apaeswq	Decreasing trend	754657	32	1995 - 2026	6.8	-0.10	7.11	-0.02	0.00
apalmwq	Decreasing trend	298361	11	2016 - 2026	6.4	-0.15	6.78	-0.05	0.05
apapcwq	No detectable trend	311516	11	2016 - 2026	6.9	-0.03	6.94	-0.01	0.76

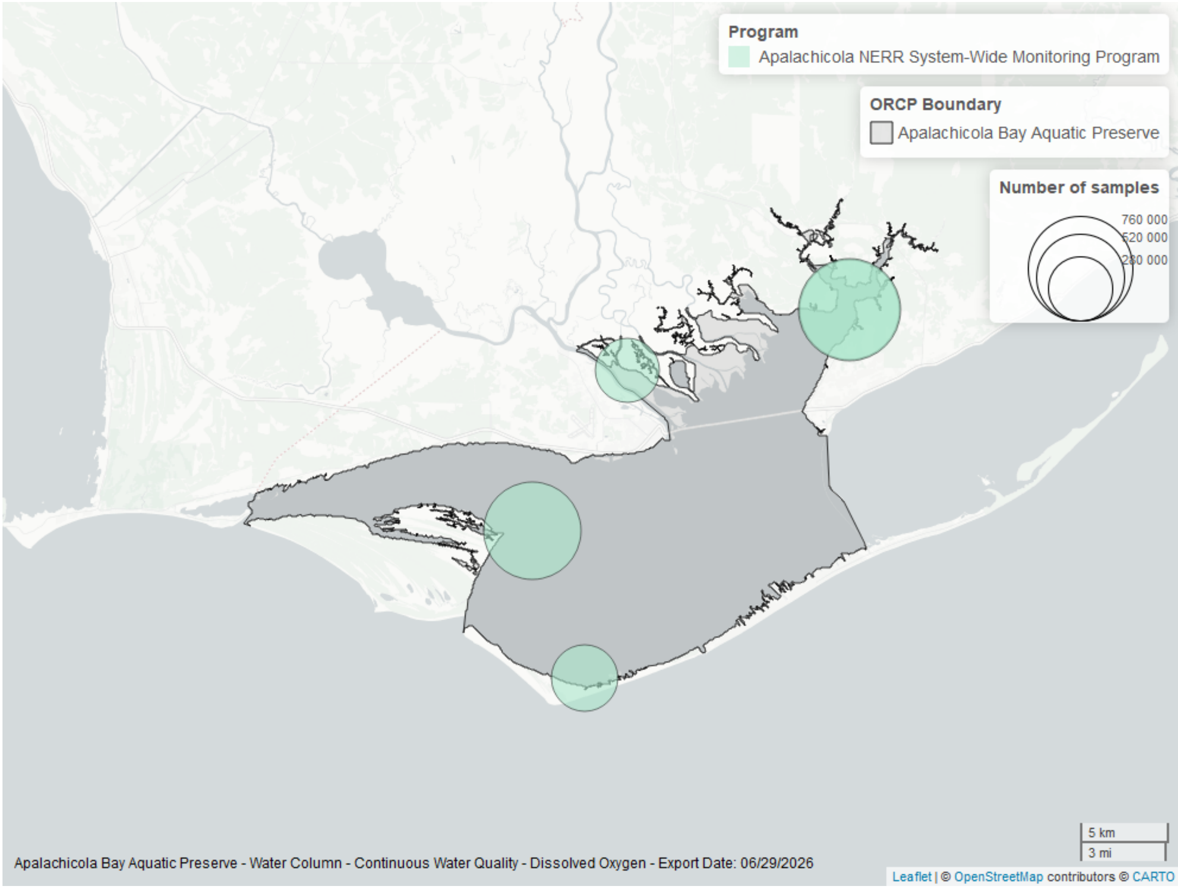

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

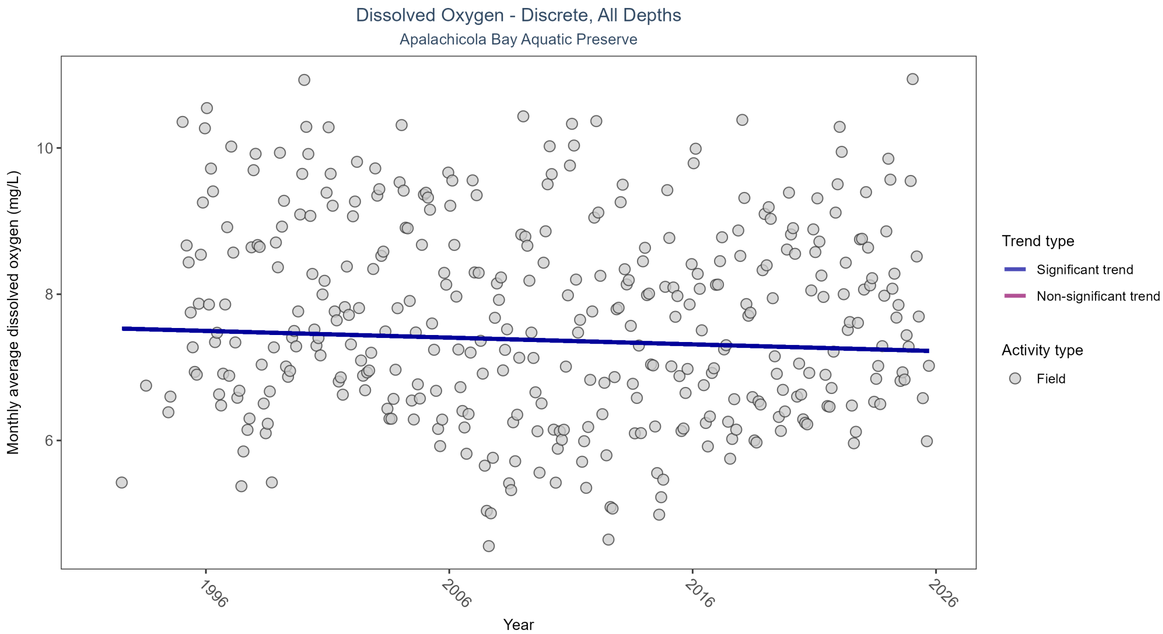

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	53625	34	1992 - 2025	7.5	-0.07801	7.53505	-0.00919	0.034

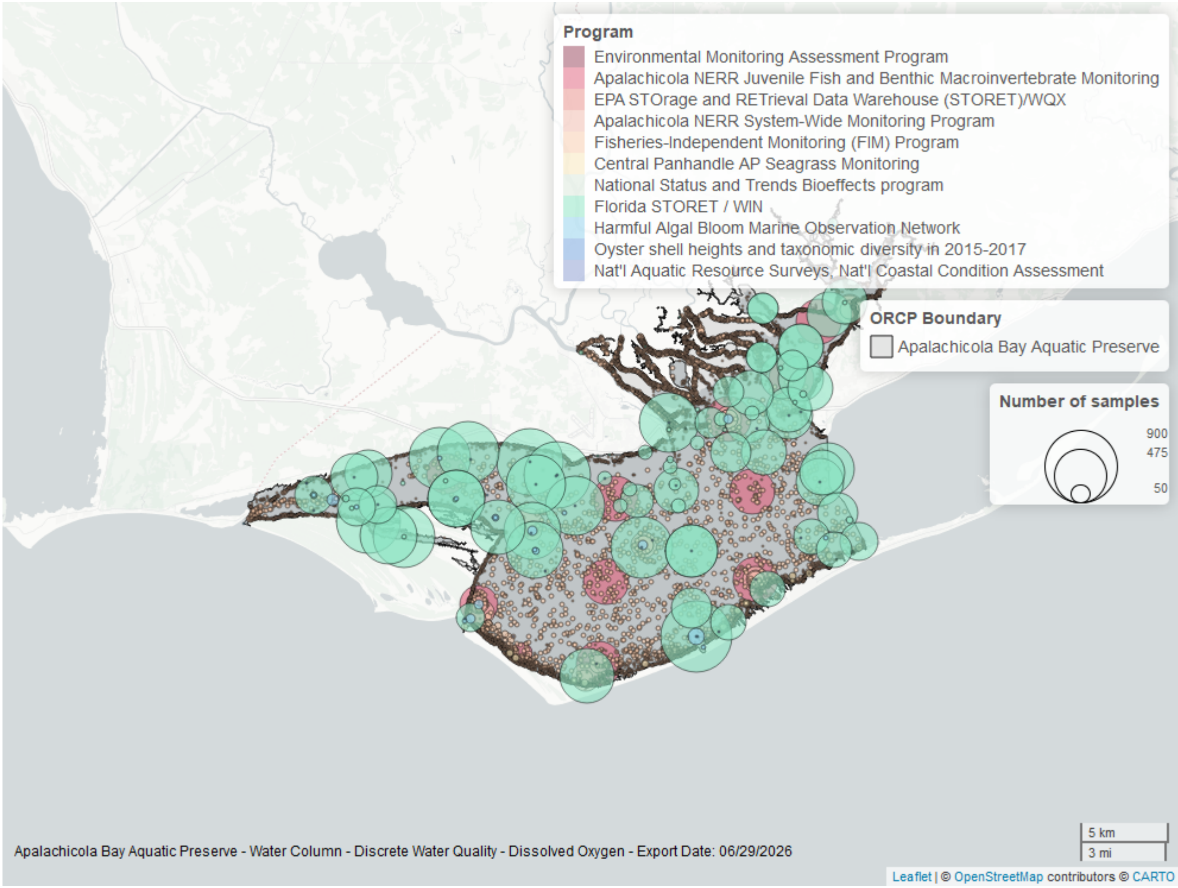

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

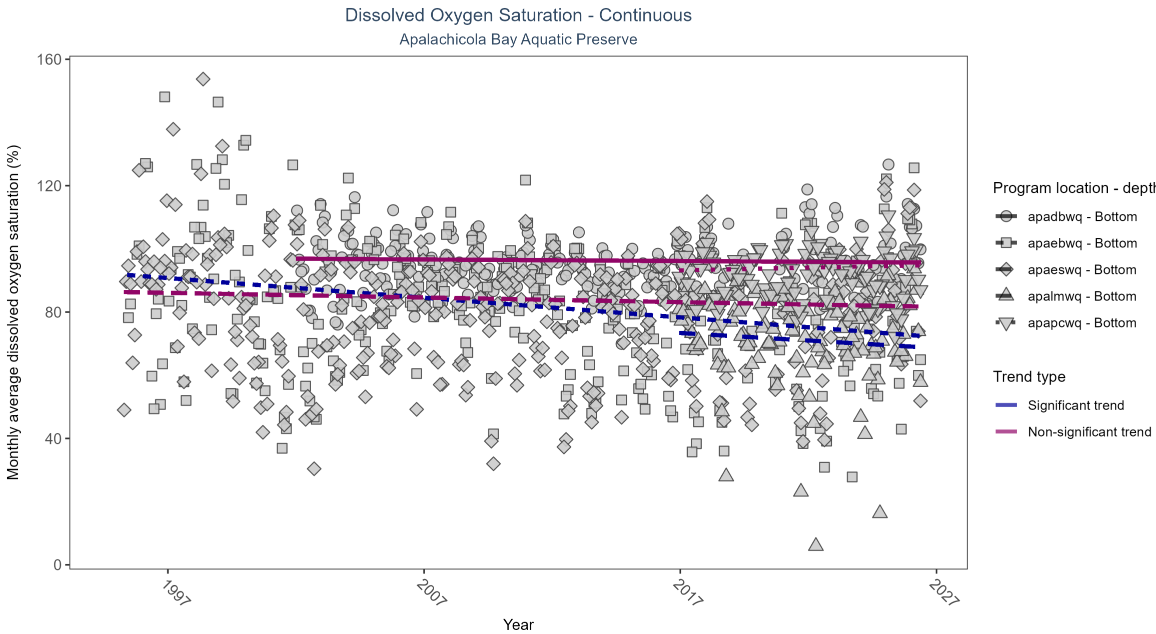

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At two program locations, monthly average dissolved oxygen saturation decreased by 0.48% per year at one site and by 0.62% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at three locations.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No detectable trend	665776	25	2002 - 2026	94.9	-0.03	96.89	-0.05	0.49
apaebwq	Decreasing trend	701762	32	1995 - 2026	84.8	-0.25	92.00	-0.62	0.00
apaeswq	No detectable trend	755867	32	1995 - 2026	84.6	-0.06	86.37	-0.15	0.11
apalmwq	Decreasing trend	298897	11	2016 - 2026	75.2	-0.15	73.85	-0.48	0.04
apapcwq	No detectable trend	314972	11	2016 - 2026	94.2	0.06	93.04	0.17	0.39

\end{table}

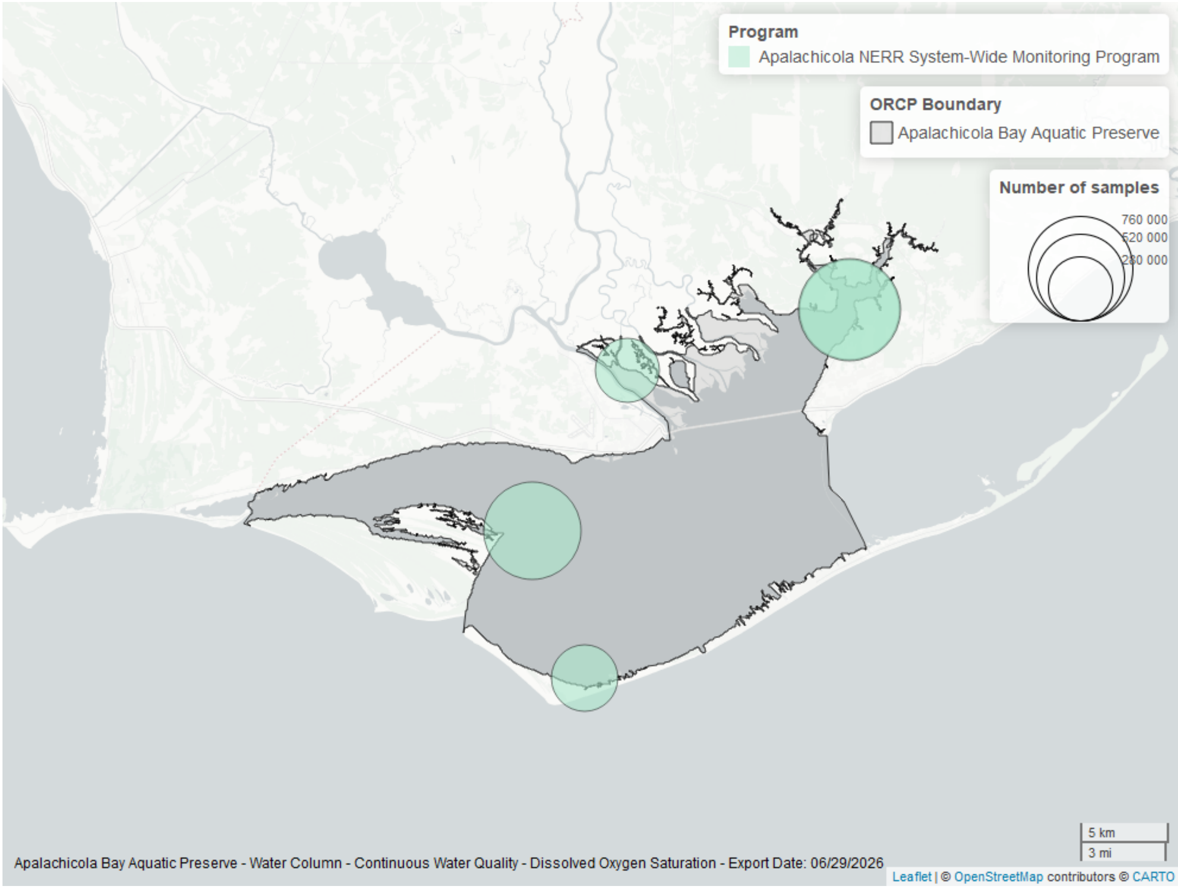

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

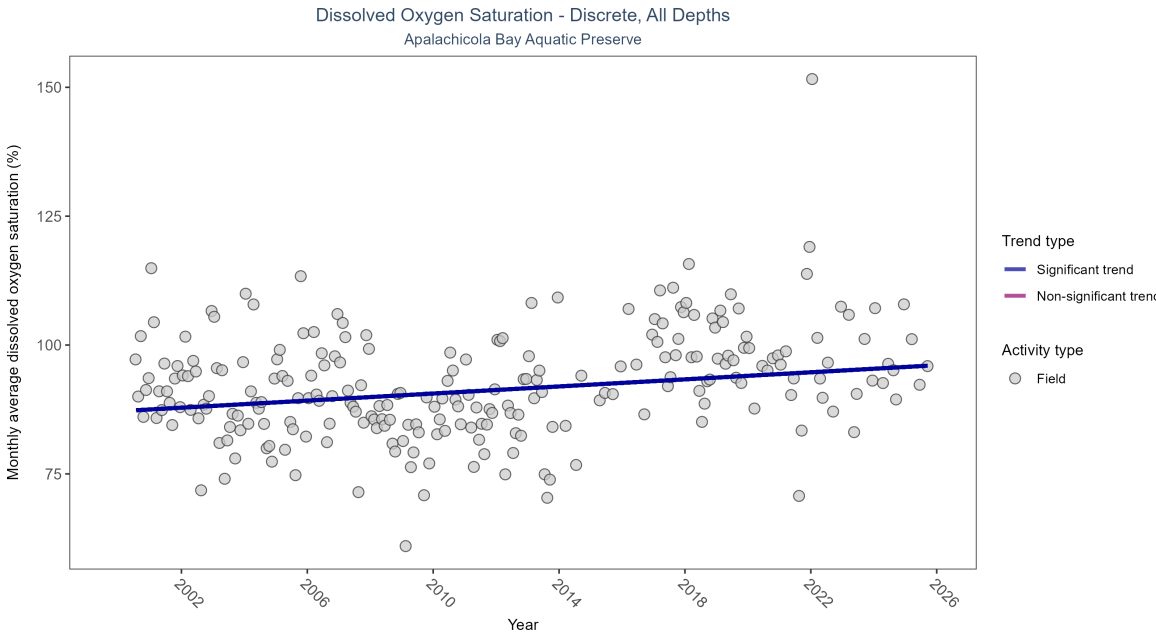

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.34% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	4593	26	2000 - 2025	92.7	0.19002	87.1556	0.34348	0

\end{table}

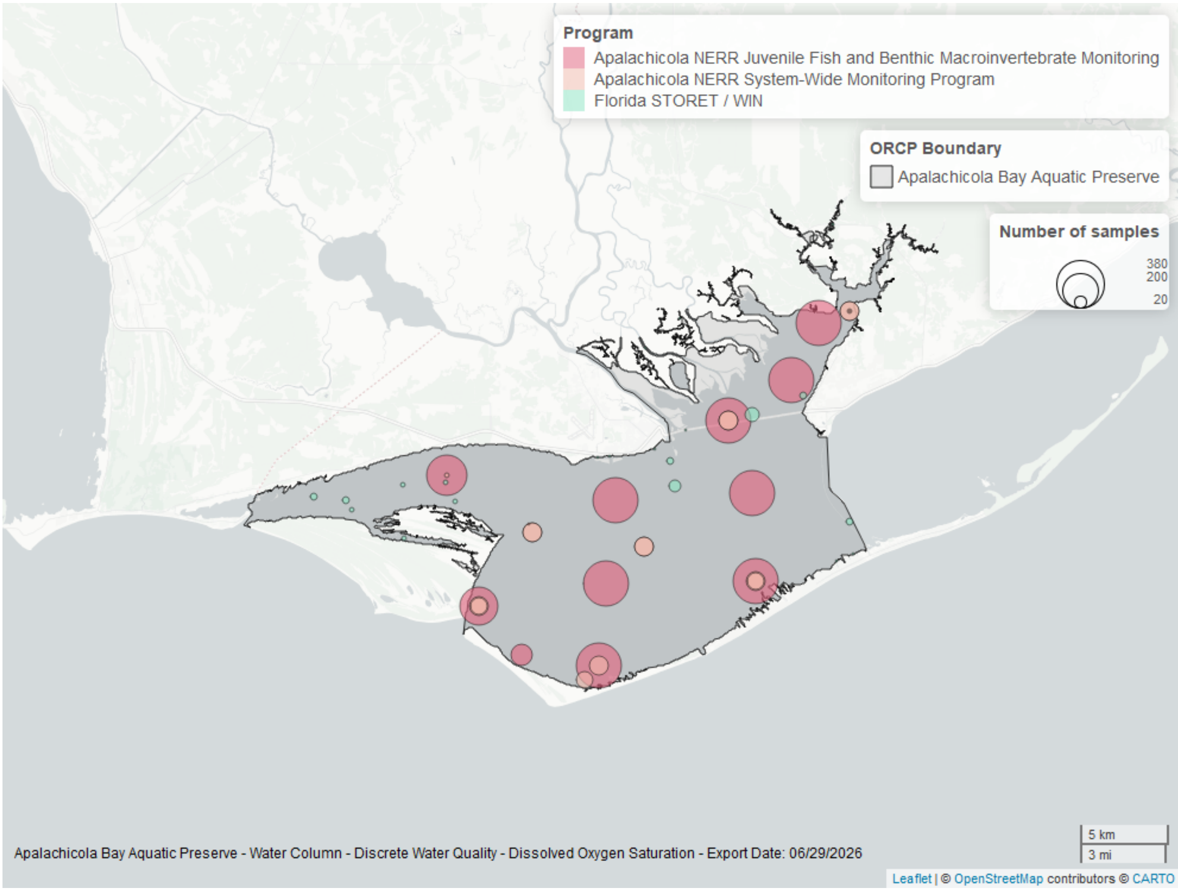

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

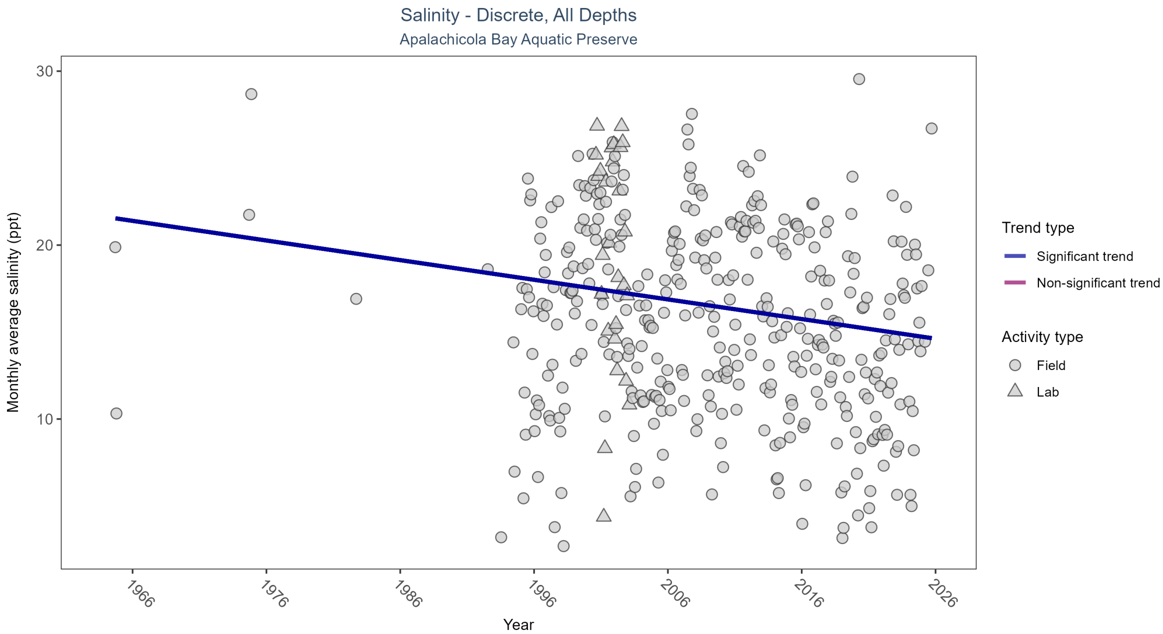

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Monthly average salinity decreased by 0.11 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	62527	37	1964 - 2025	15.8	-0.15607	21.61929	-0.11291	0

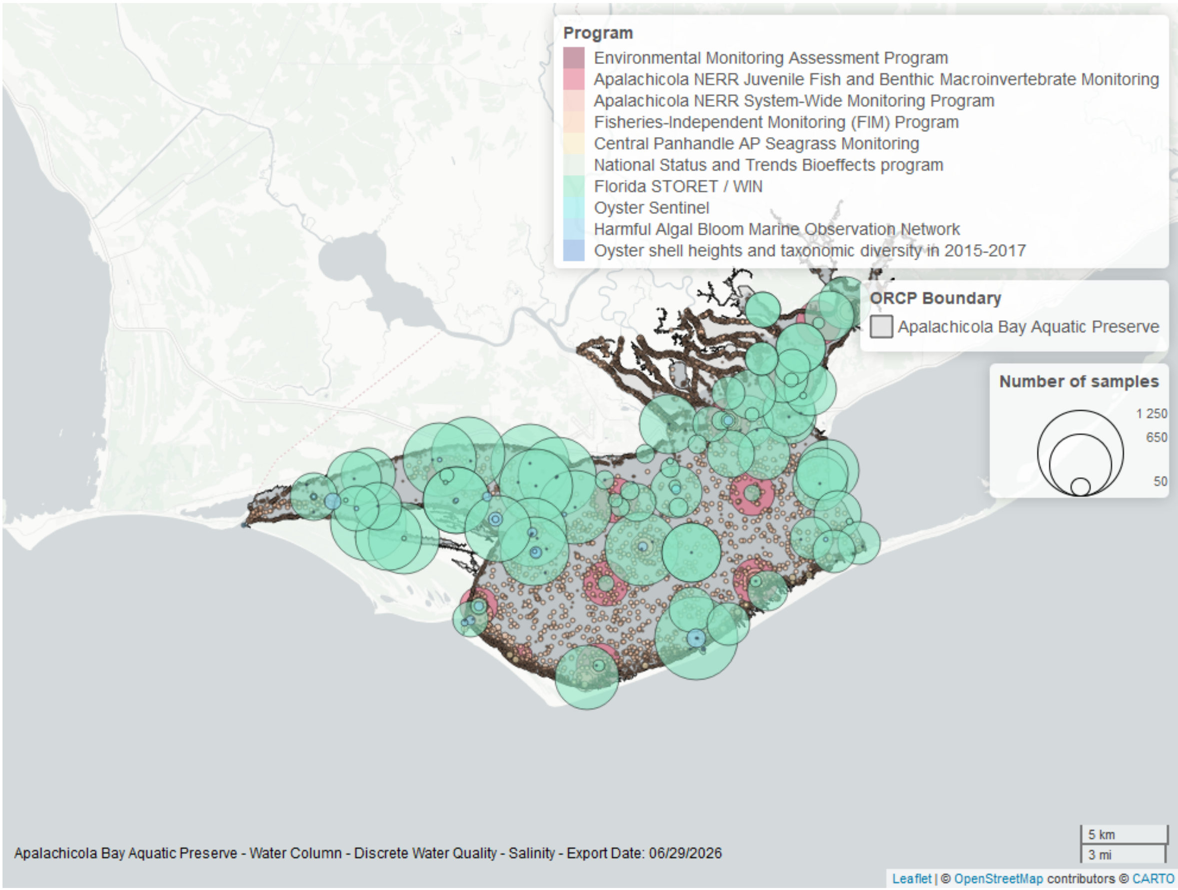

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

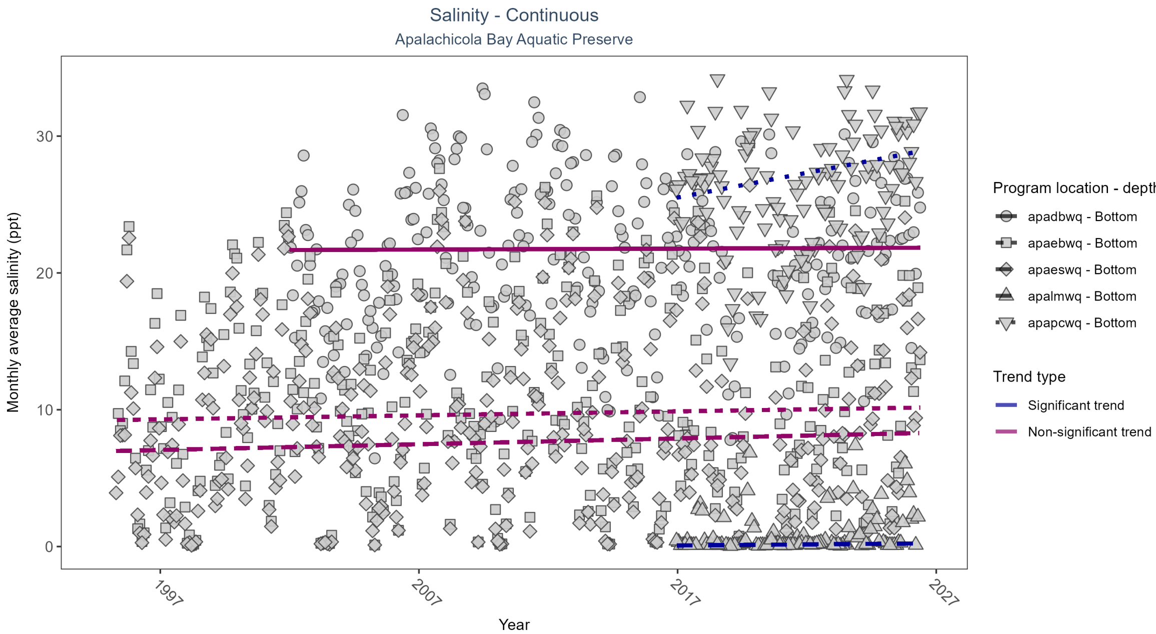

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At two program locations, monthly average salinity increased by 0.02 ppt per year at one site and by 0.36 ppt per year at the other. No detectable change in monthly average salinity was observed at three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No detectable trend	660145	25	2002 - 2026	22.3	0.01	21.67	0.01	0.85
apaebwq	No detectable trend	792228	32	1995 - 2026	10.1	0.05	9.24	0.03	0.17
apaeswq	No detectable trend	800886	32	1995 - 2026	7.6	0.07	6.97	0.04	0.07
apalmwq	Increasing trend	312951	11	2016 - 2026	0.1	0.28	0.06	0.02	0.00
apapcwq	Increasing trend	312976	11	2016 - 2026	27.3	0.19	25.15	0.36	0.01

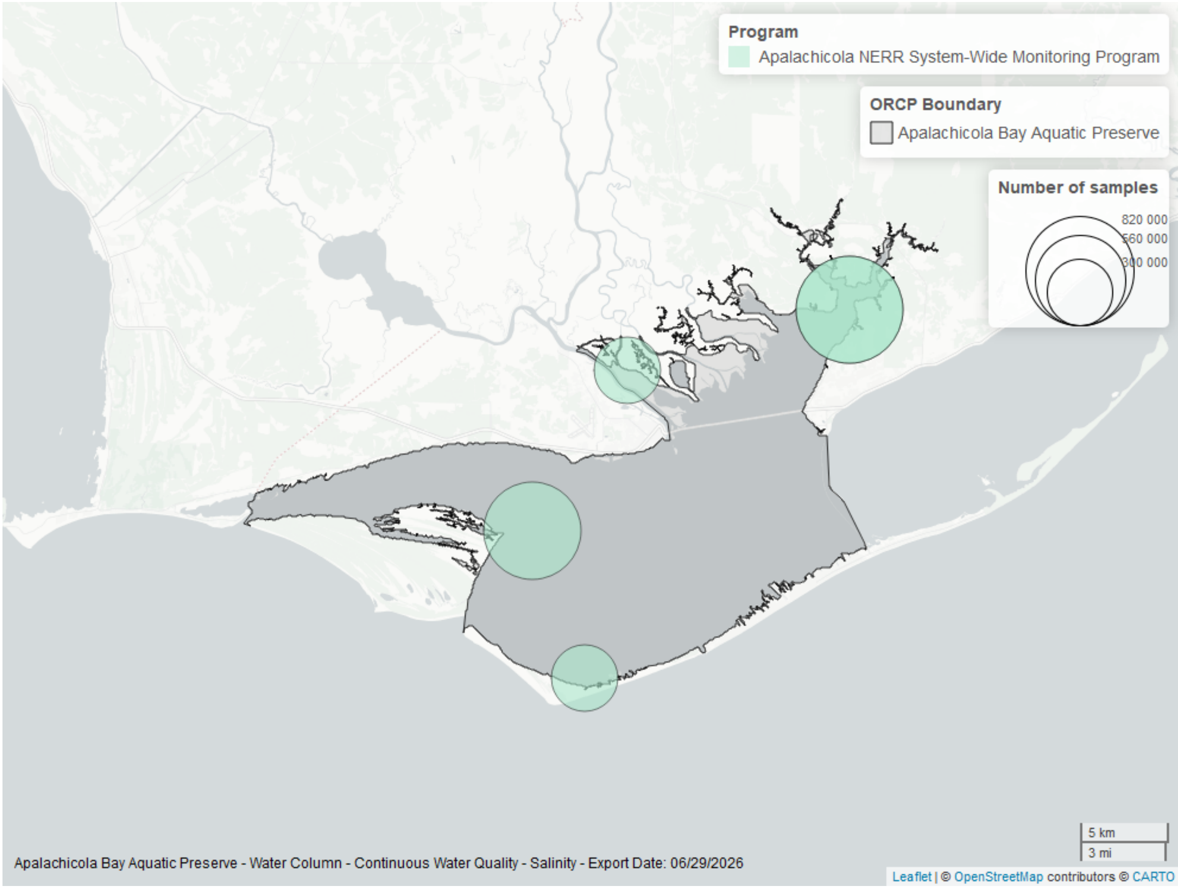

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

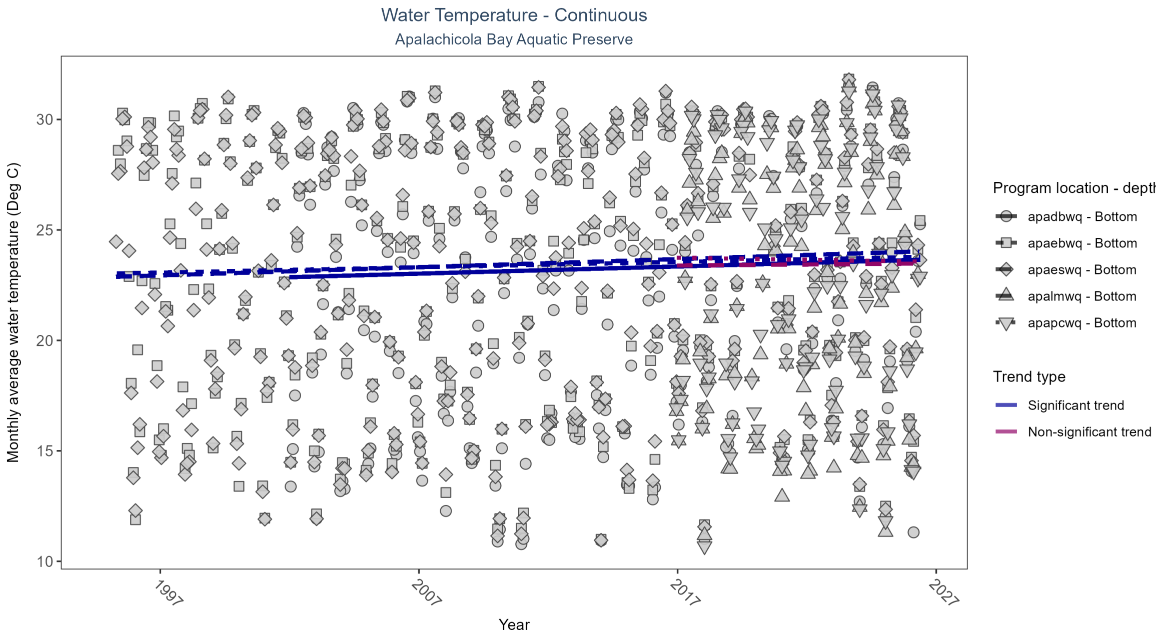

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At three program locations, monthly average water temperature increased between 0.02 and 0.04Deg C per year. No detectable change in monthly average water temperature was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Increasing trend	687997	25	2002 - 2026	23.3	0.16	22.86	0.03	0.00
apaebwq	Increasing trend	807299	32	1995 - 2026	24.1	0.16	23.02	0.02	0.00
apaeswq	Increasing trend	810119	32	1995 - 2026	24.1	0.20	22.87	0.04	0.00
apalmwq	No detectable trend	315207	11	2016 - 2026	22.5	0.04	23.38	0.01	0.69
apapcwq	No detectable trend	317938	11	2016 - 2026	22.9	-0.02	23.76	-0.02	0.80

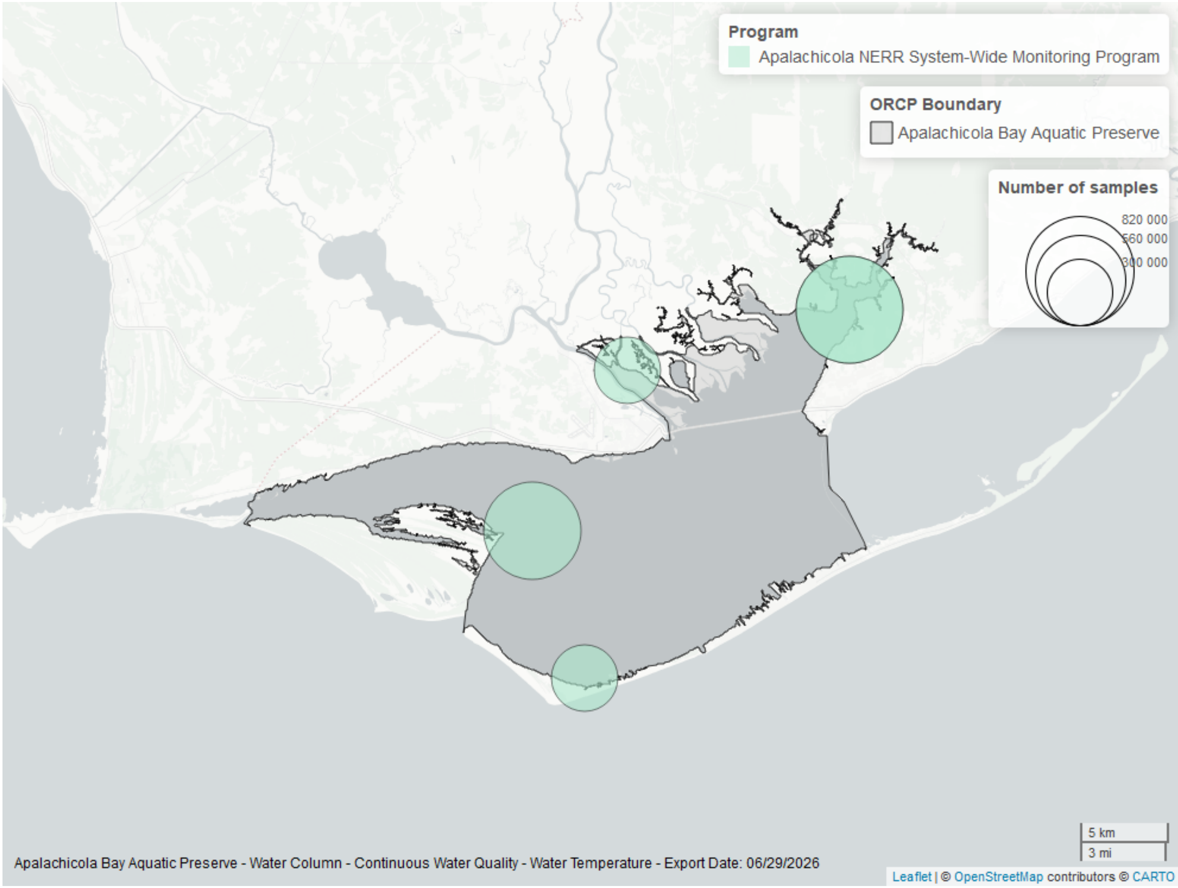

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

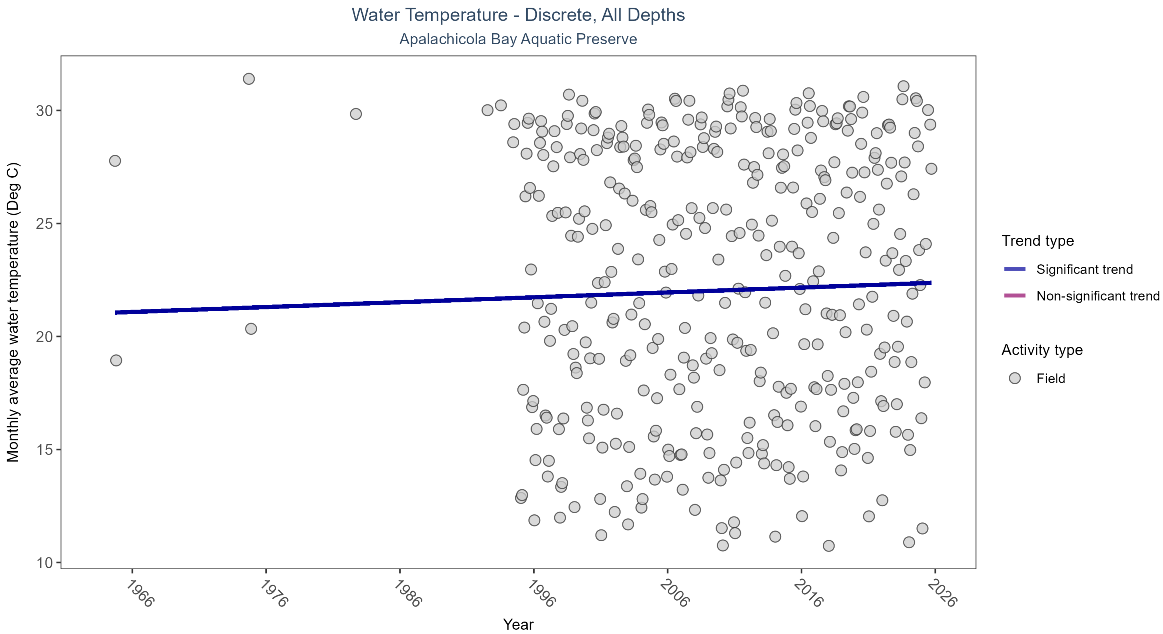

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 8: Monthly average water temperature increased by 0.02Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	62790	37	1964 - 2025	24	0.10812	21.04235	0.02157	0.0035

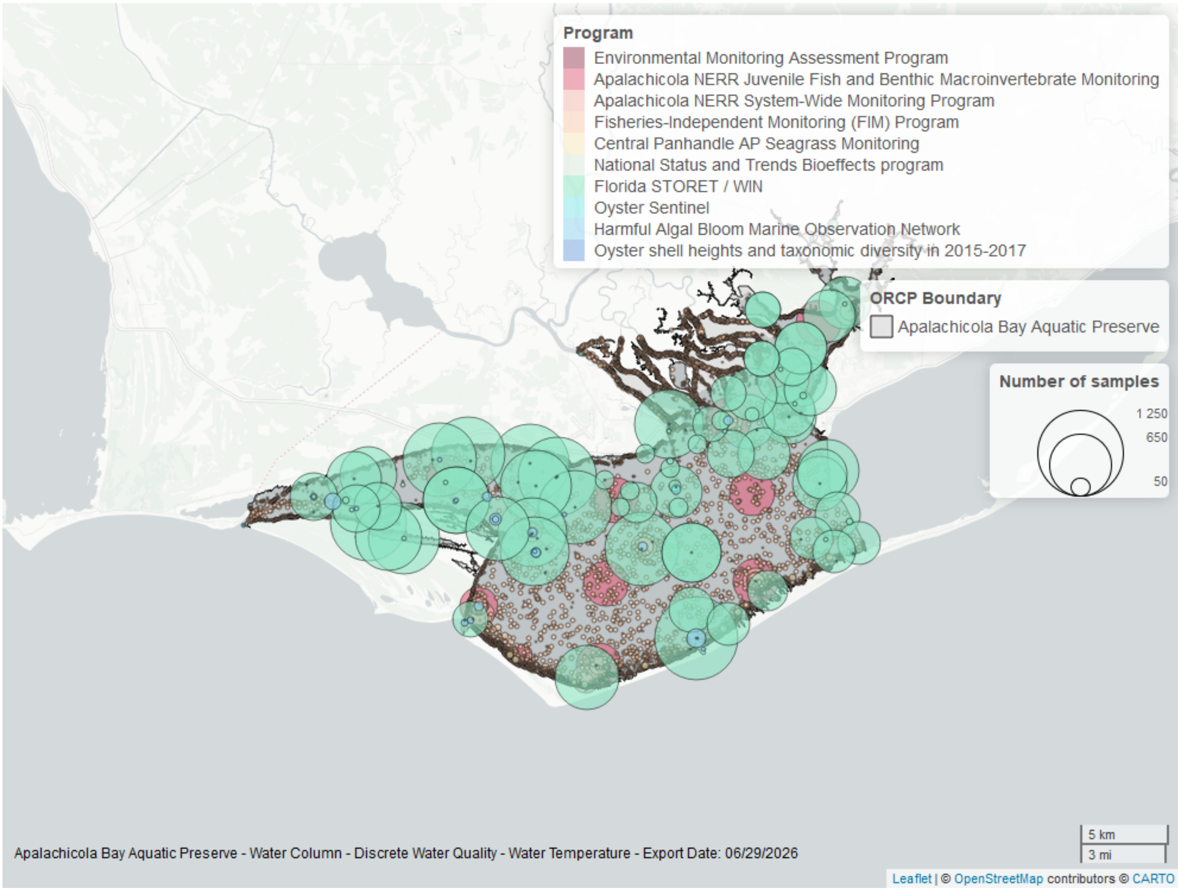

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

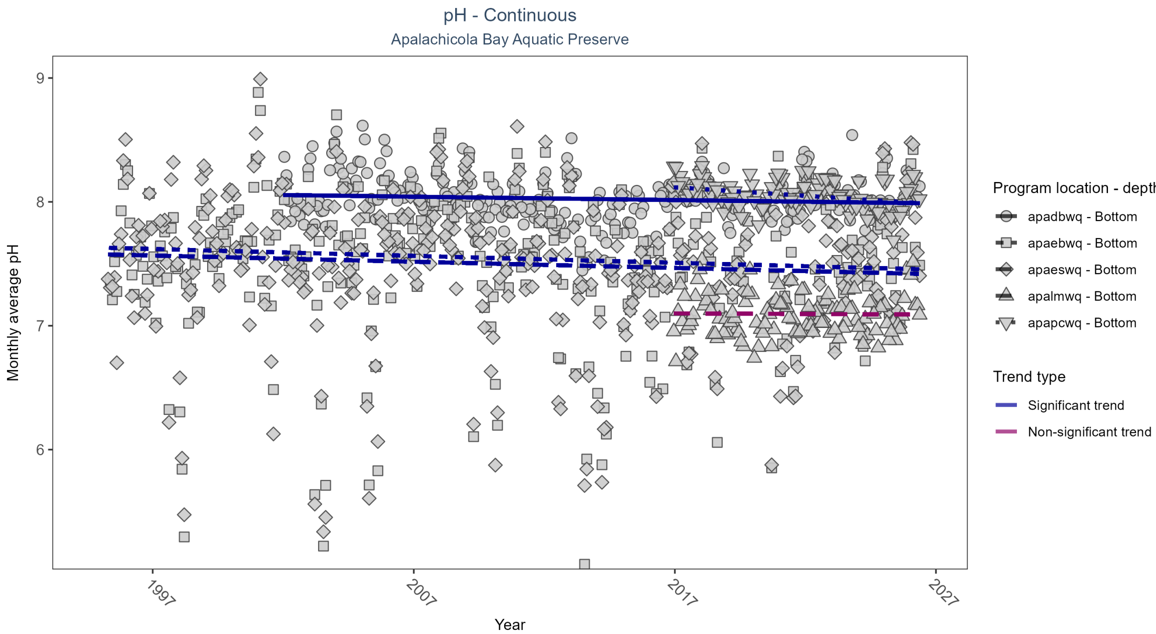

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At four program locations, monthly average pH decreased between less than 0.01 and 0.01 pH units per year. No detectable change in monthly average pH was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Decreasing trend	645135	25	2002 - 2026	8.0	-0.10	8.06	0.00	0.02
apaebwq	Decreasing trend	762842	32	1995 - 2026	7.6	-0.10	7.63	-0.01	0.01
apaeswq	Decreasing trend	755459	32	1995 - 2026	7.5	-0.09	7.58	-0.01	0.02
apalmwq	No detectable trend	306501	11	2016 - 2026	7.1	0.00	7.10	0.00	0.93
apapcwq	Decreasing trend	312076	11	2016 - 2026	8.1	-0.30	8.13	-0.01	0.00

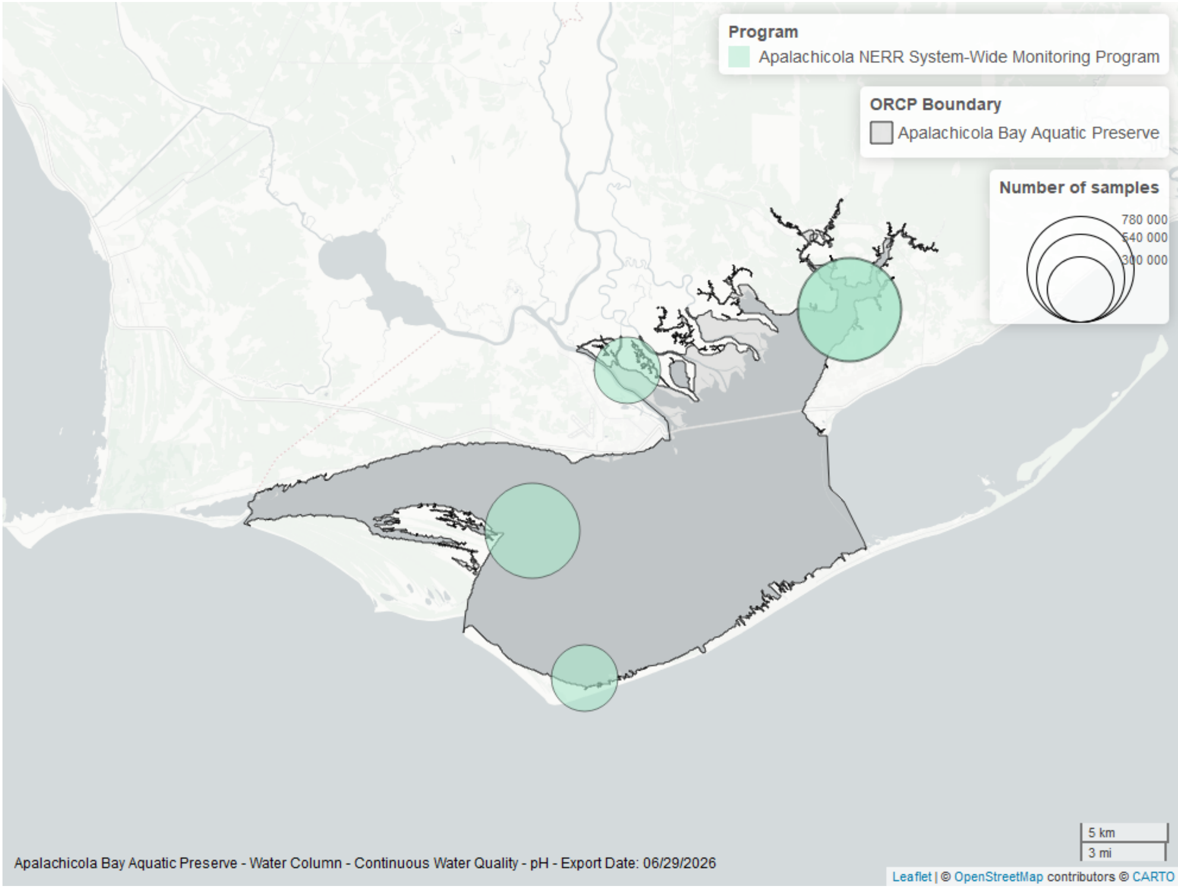

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

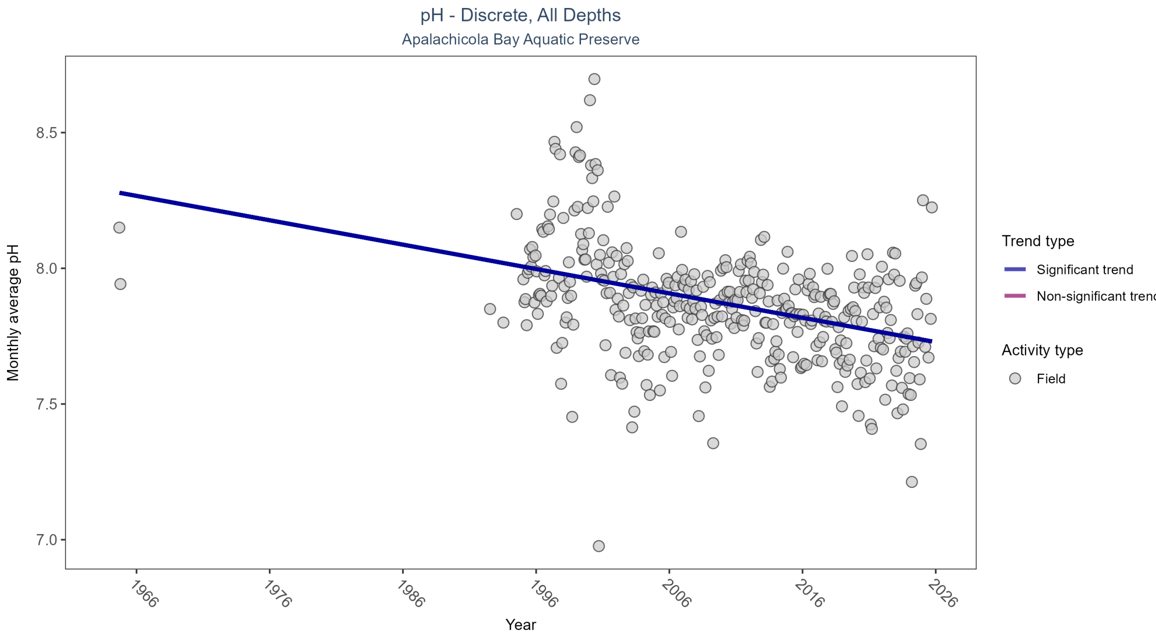

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 10: Monthly average pH decreased by 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	42722	35	1964 - 2025	8	-0.29544	8.28476	-0.00898	0

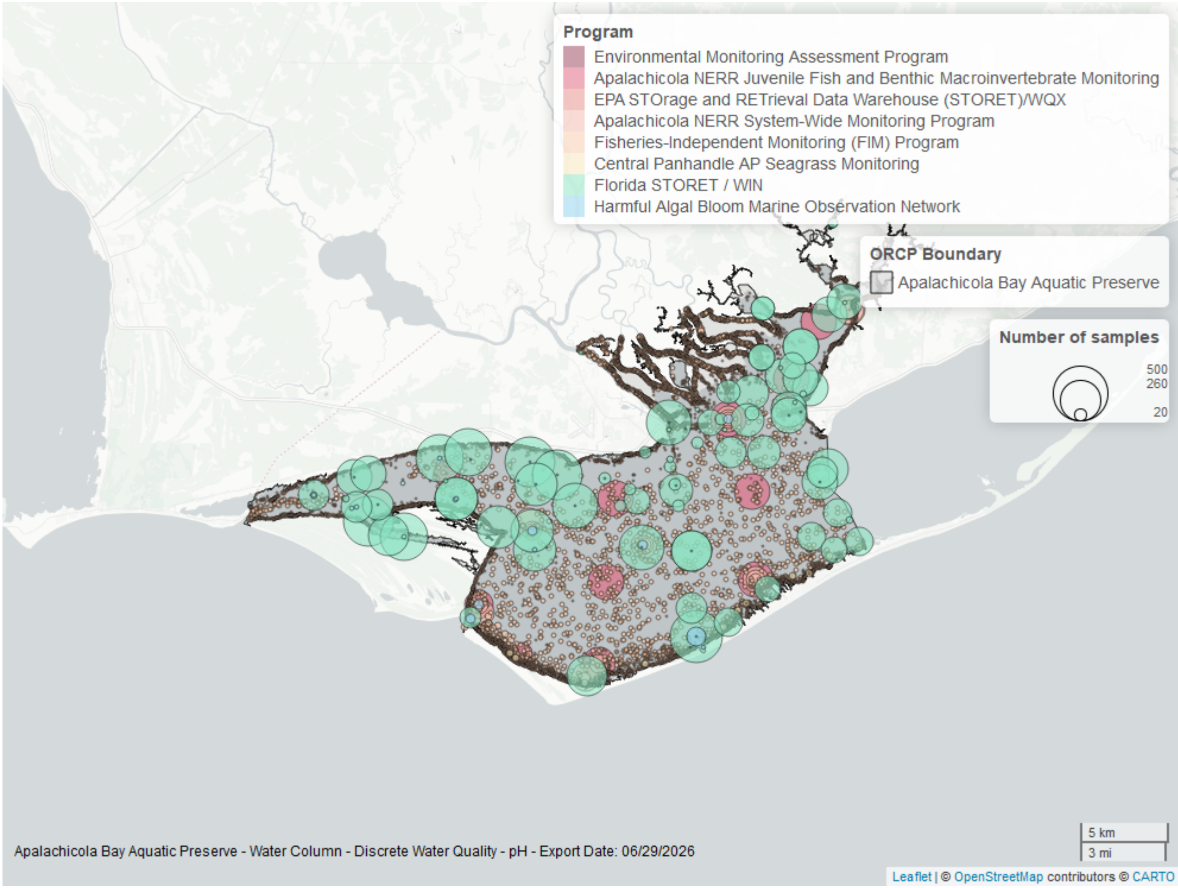

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

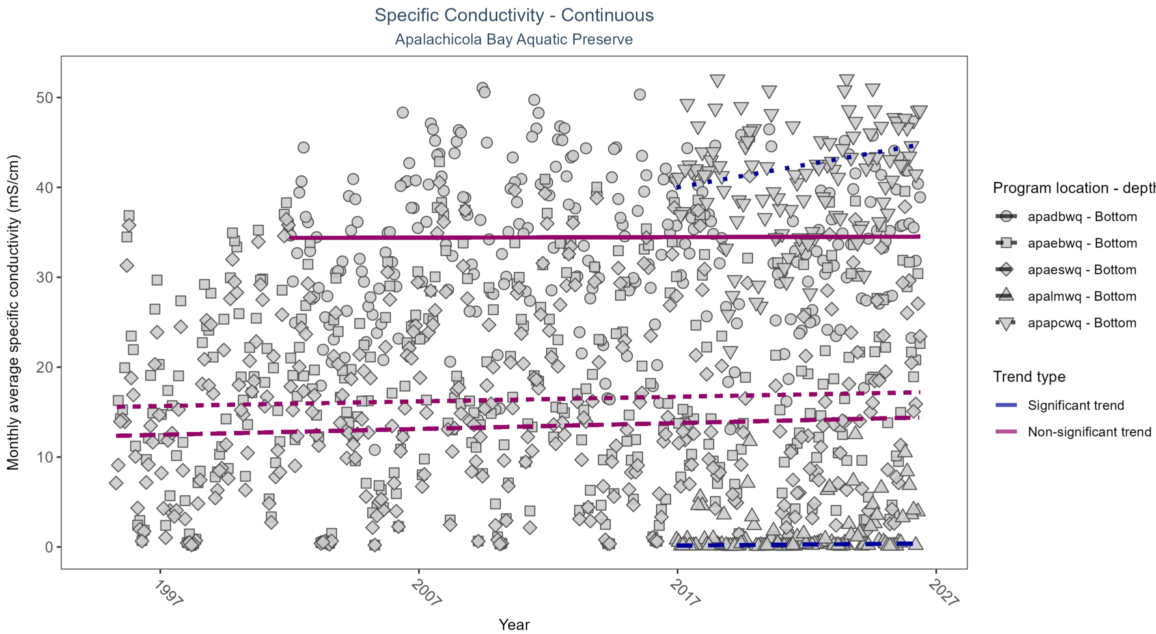

Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At two program locations, monthly average specific conductivity increased by 0.02 mS/cm per year at one site and by 0.51 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No detectable trend	660231	25	2002 - 2026	35.38	0.01	34.38	0.01	0.92
apaebwq	No detectable trend	792357	32	1995 - 2026	17.12	0.05	15.57	0.05	0.16
apaeswq	No detectable trend	801369	32	1995 - 2026	13.27	0.07	12.34	0.07	0.08
apalmwq	Increasing trend	312951	11	2016 - 2026	0.15	0.27	0.13	0.02	0.00
apapcwq	Increasing trend	312978	11	2016 - 2026	42.46	0.20	39.50	0.51	0.01

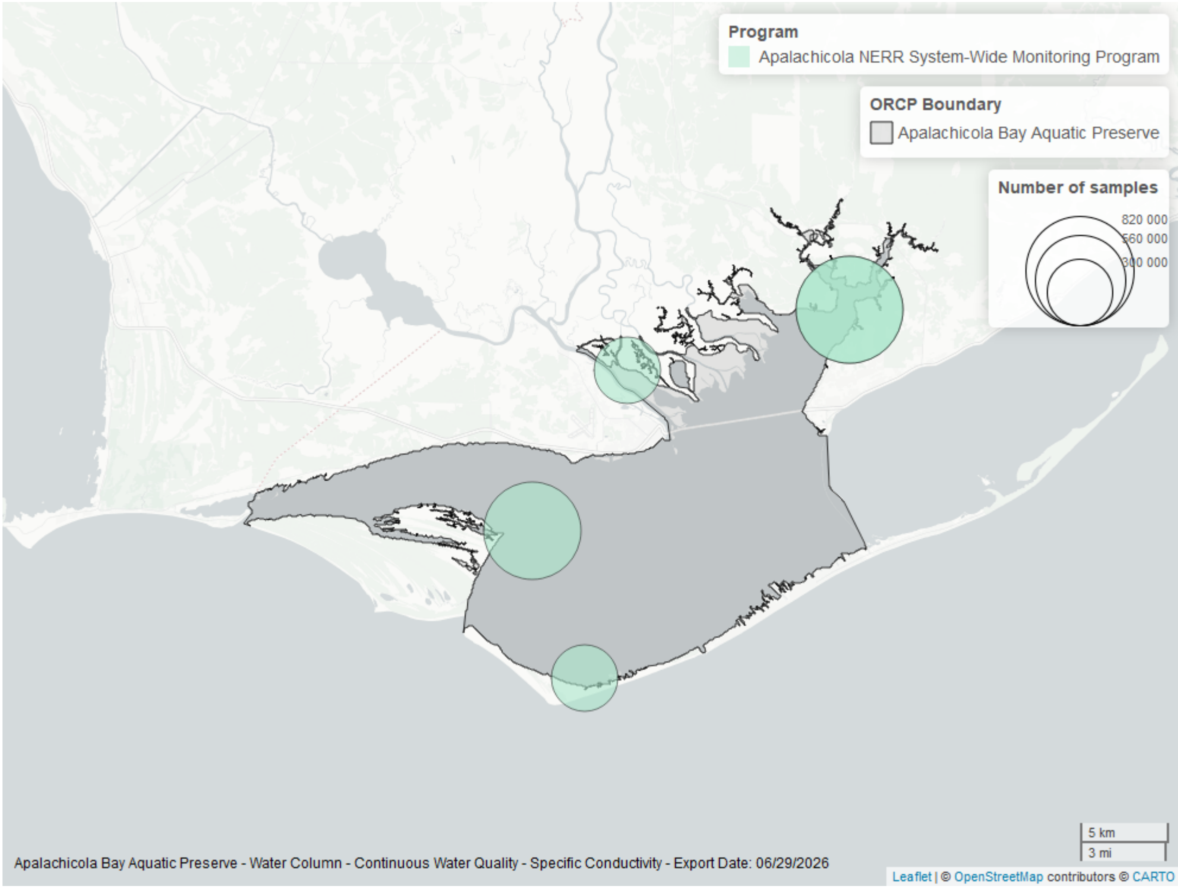

Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

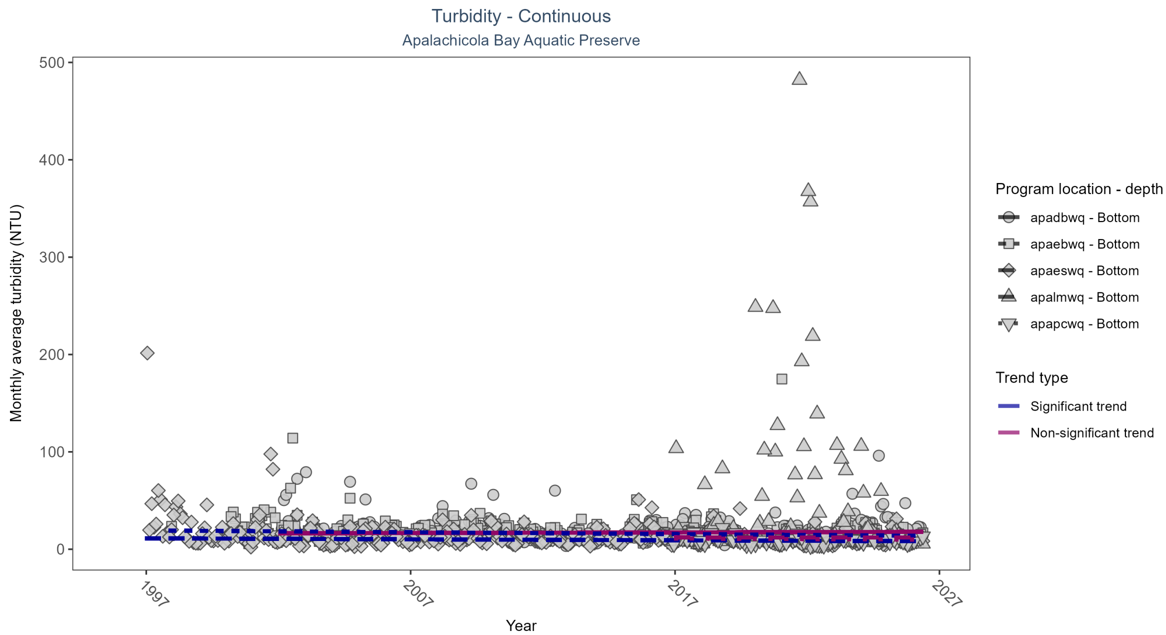

Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At two program locations, monthly average turbidity decreased by 0.1 NTU per year at one site and by 0.19 NTU per year at the other. No detectable change in monthly average turbidity was observed at three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No detectable trend	647066	25	2002 - 2026	10	0.06	16.34	0.06	0.20
apaebwq	Decreasing trend	680779	28	1997 - 2026	13	-0.19	19.33	-0.19	0.00
apaeswq	Decreasing trend	744747	31	1996 - 2026	9	-0.14	11.31	-0.10	0.00
apalmwq	No detectable trend	284580	11	2016 - 2026	11	0.00	12.15	-0.01	1.00
apapcwq	No detectable trend	301927	11	2016 - 2026	7	-0.07	11.10	-0.11	0.36

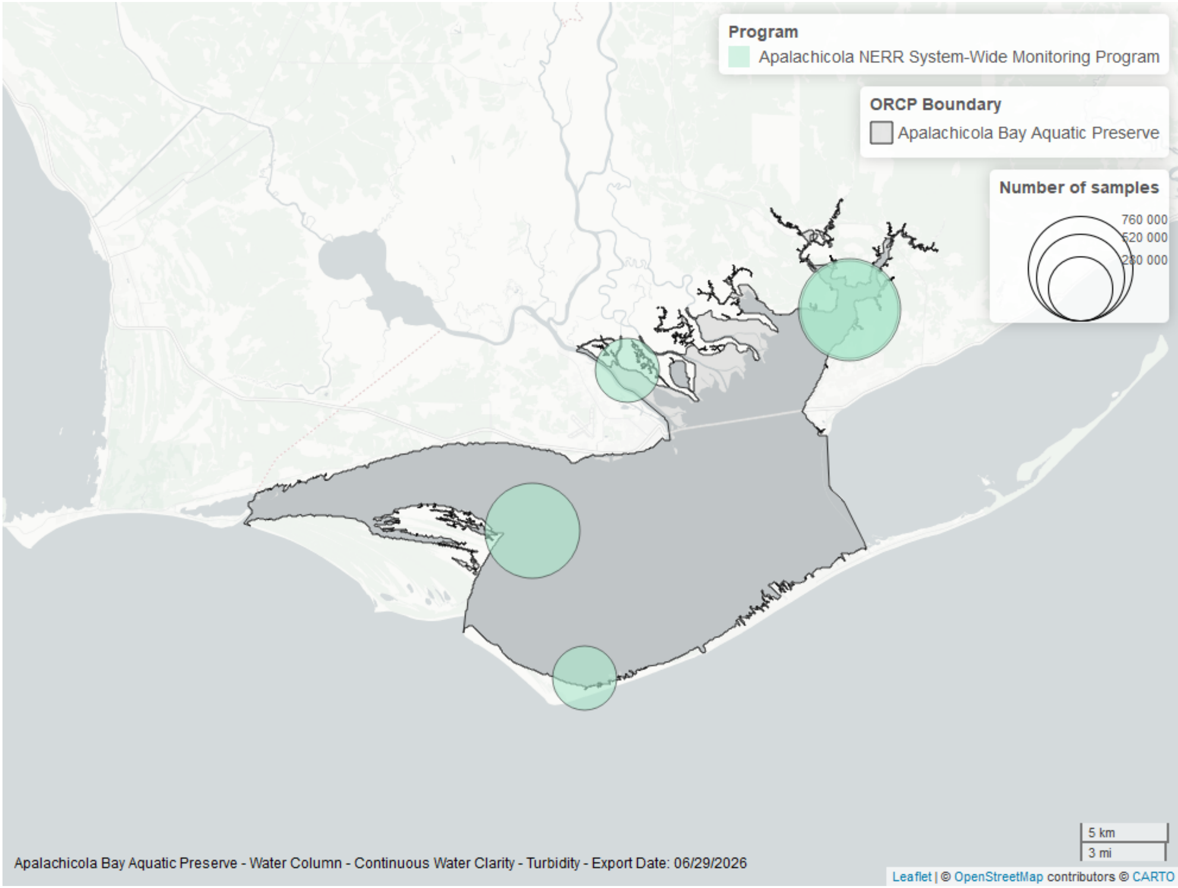

Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

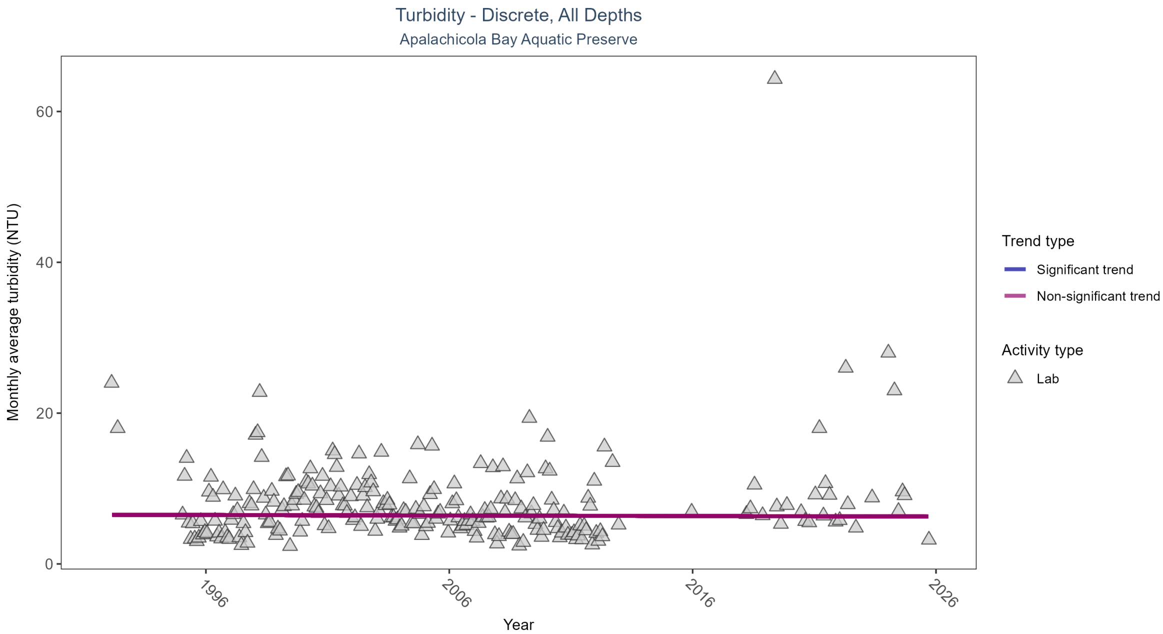

Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 13: Turbidity showed no detectable trend between 1992 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	15924	28	1992 - 2025	5.6	-0.01893	6.51601	-0.00677	0.7401

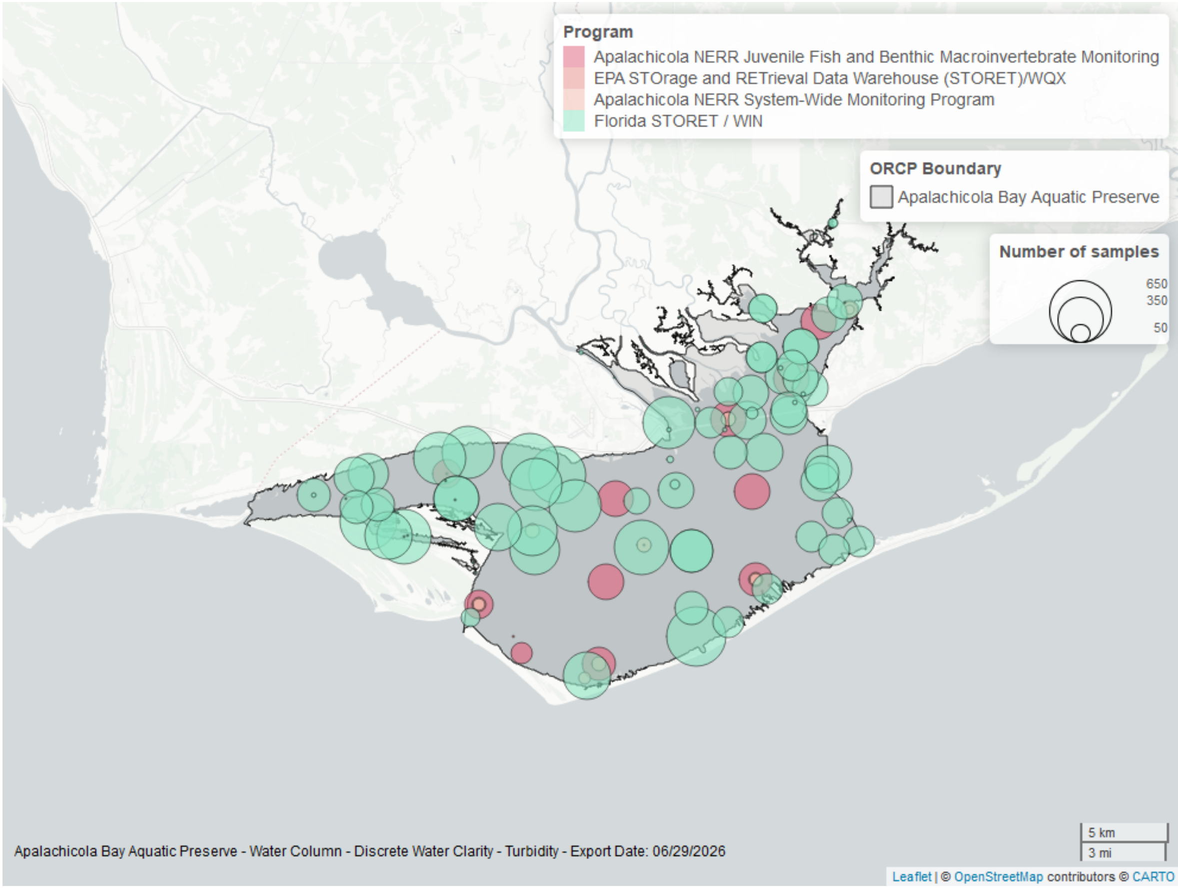

Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

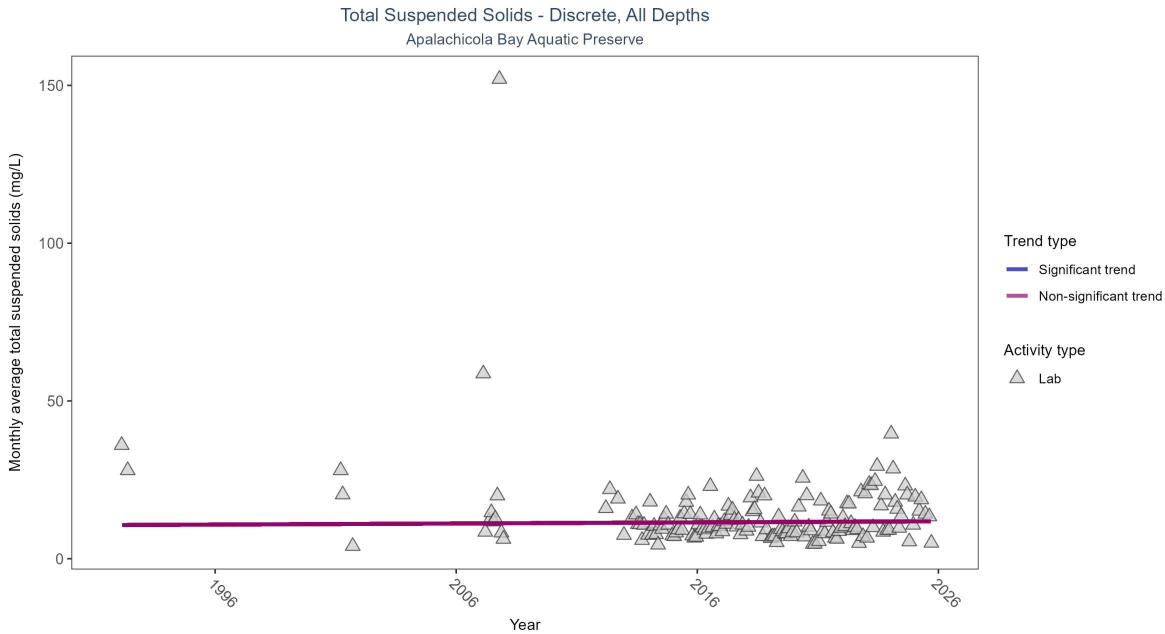

Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Total suspended solids showed no detectable trend between 1992 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2579	17	1992 - 2025	10	0.02597	10.67726	0.0349	0.8229

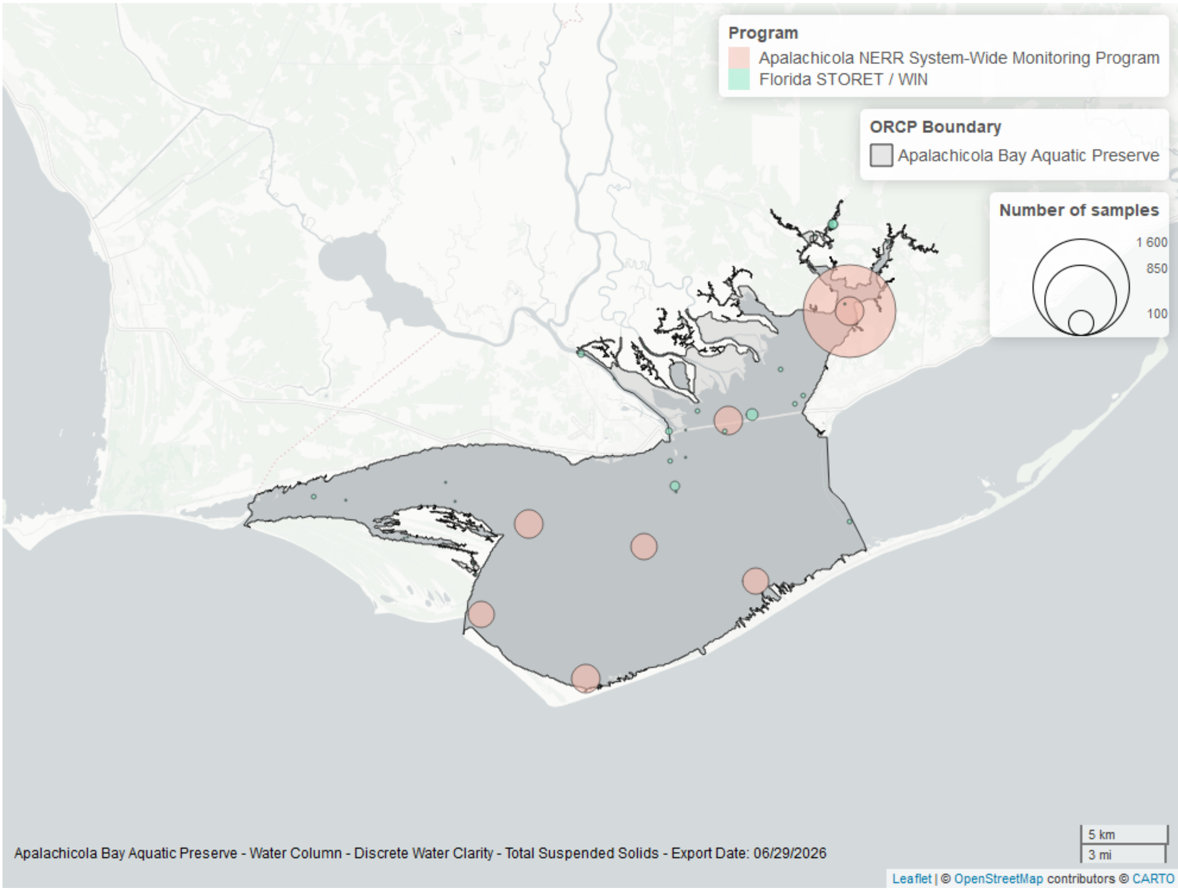

Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

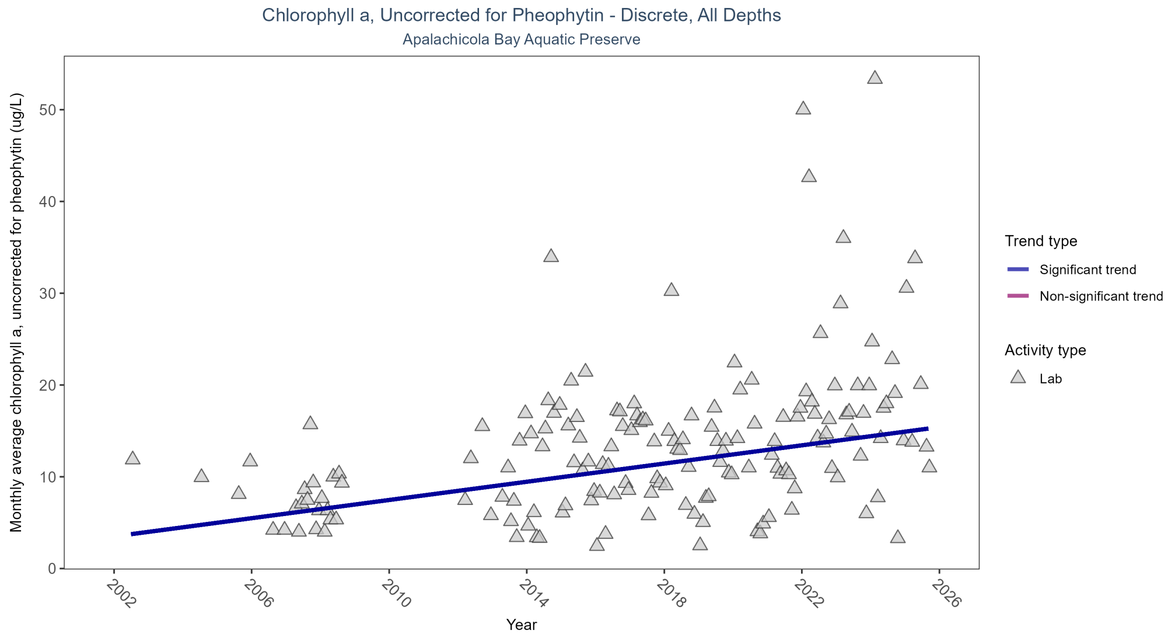

Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 15: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.5 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2568	20	2002 - 2025	12	0.35787	3.50023	0.4963	0

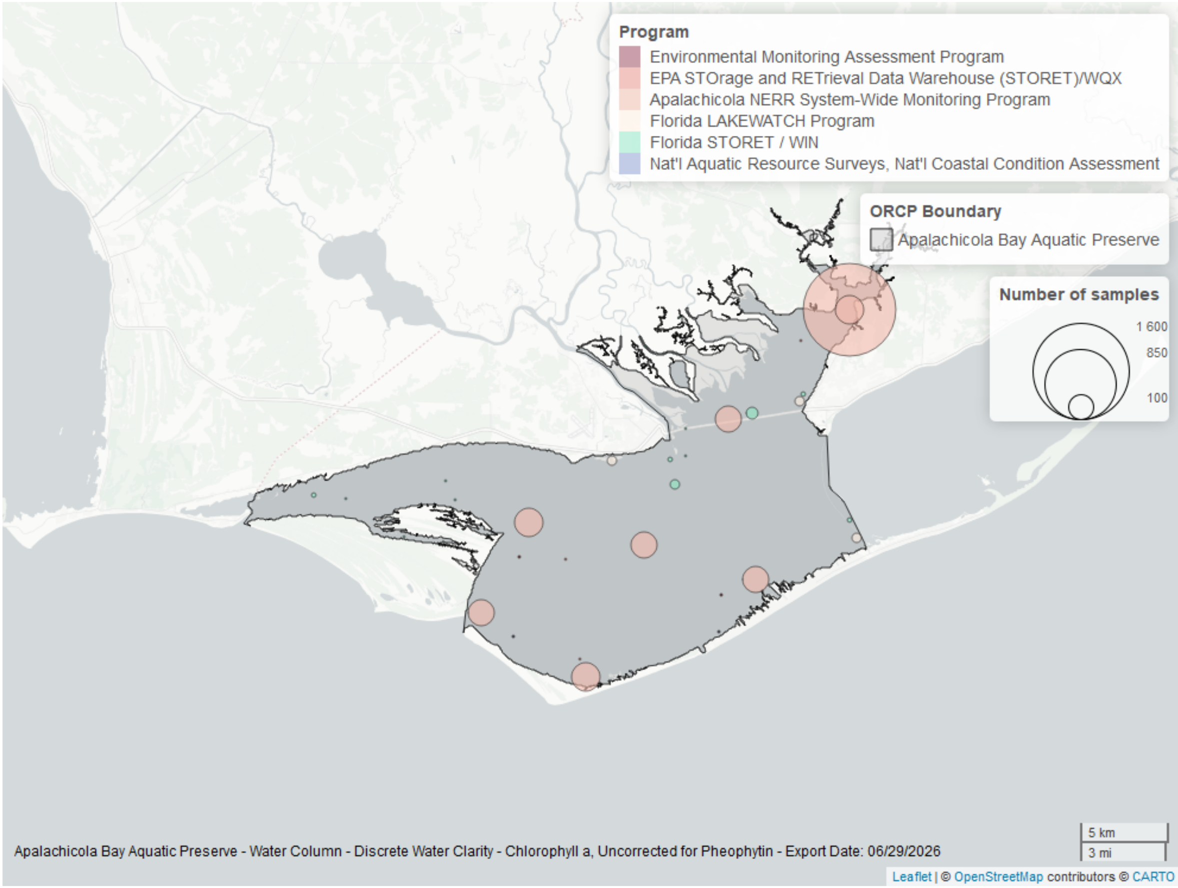

Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

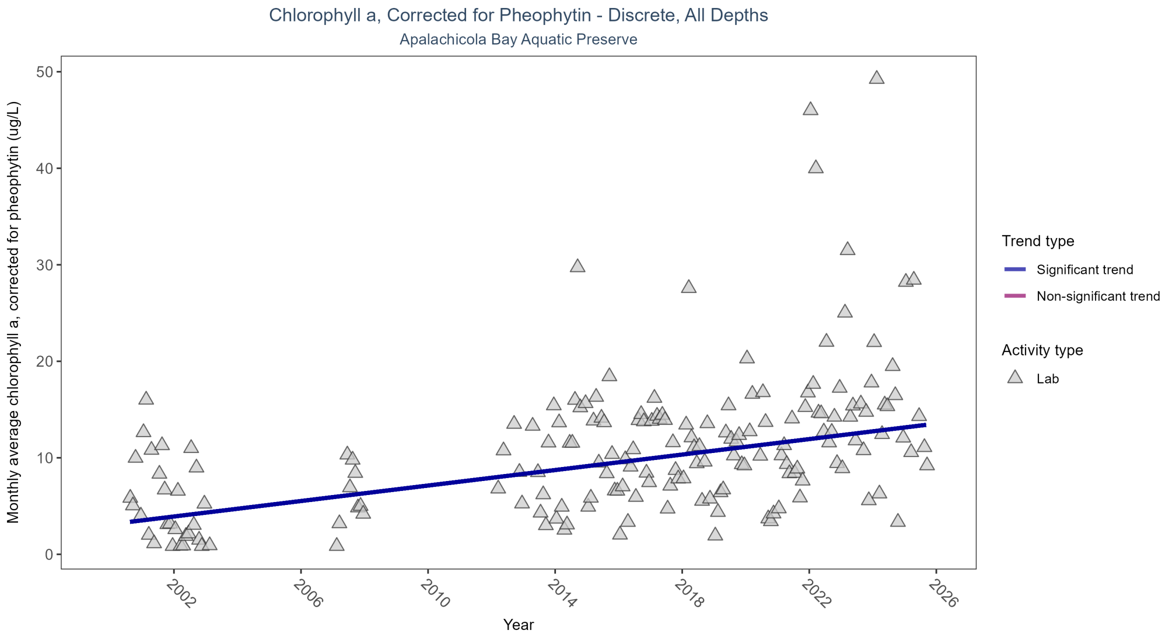

Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, corrected for pheophytin, increased by 0.4 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2941	19	2000 - 2025	9.3	0.40115	3.11584	0.40104	0

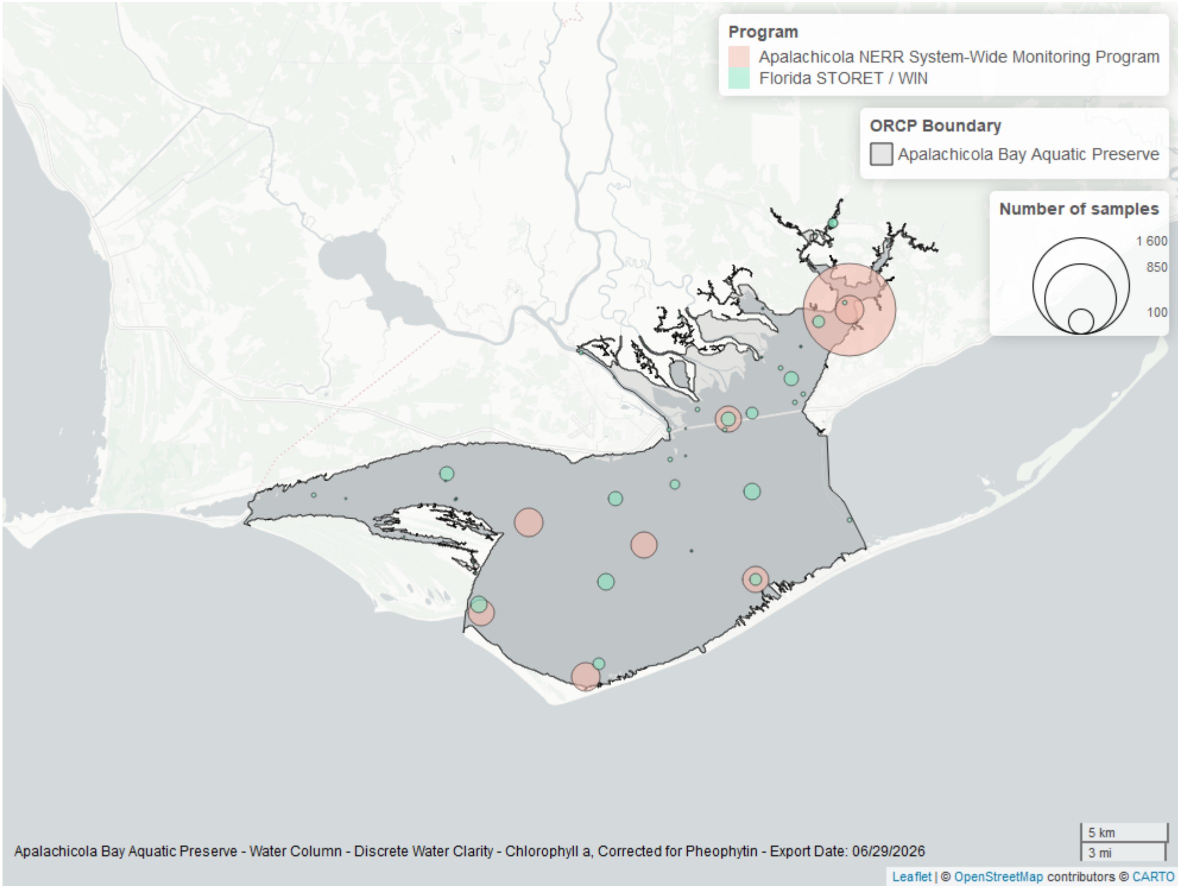

Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

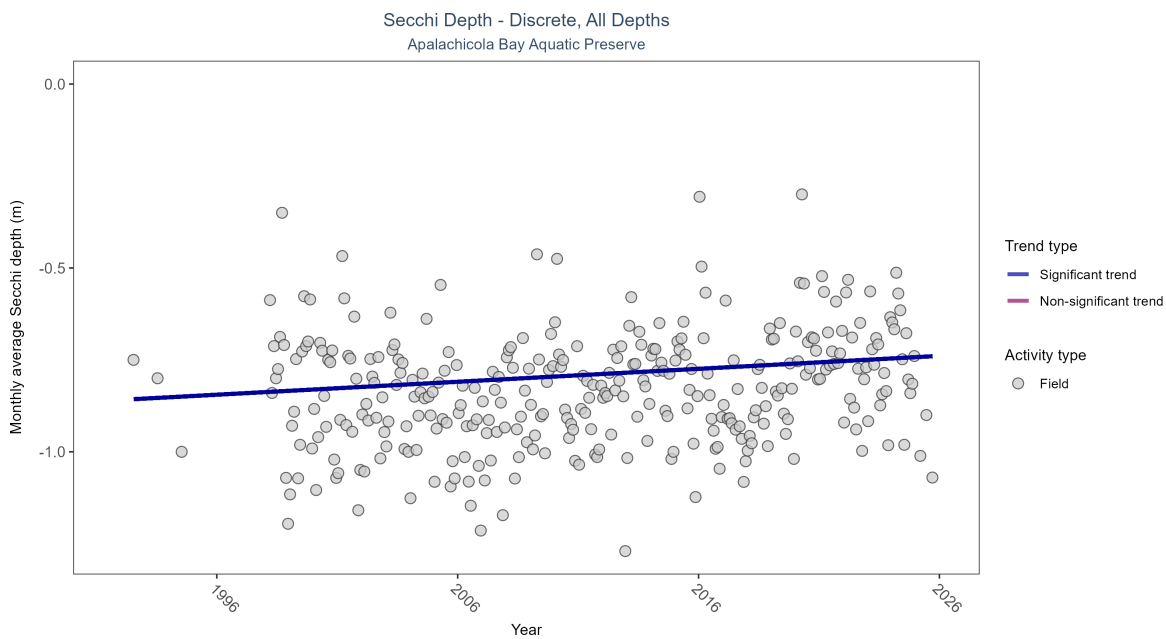

Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 17: Monthly average Secchi depth became shallower by less than 0.01 m per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	28513	31	1992 - 2025	-0.8	0.14687	-0.8588	0.00352	0.0003

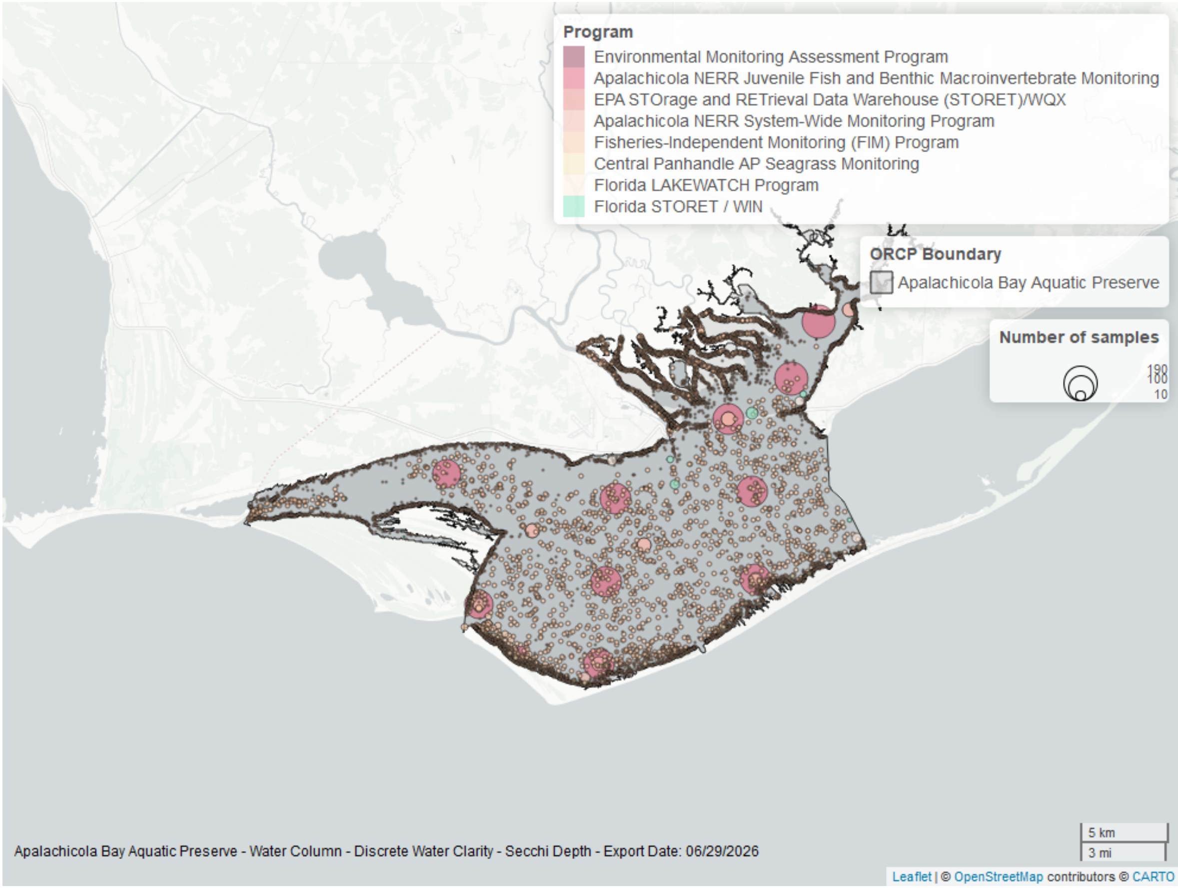

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

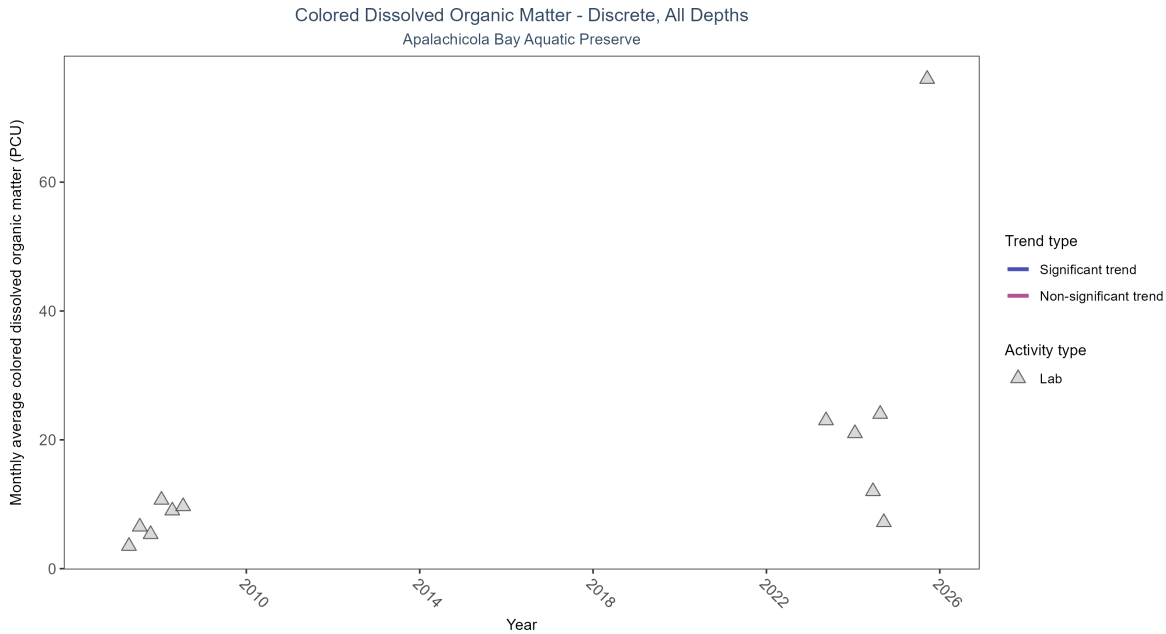

Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 18: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	22	5	2007 - 2025	7.6	-	-	-	-

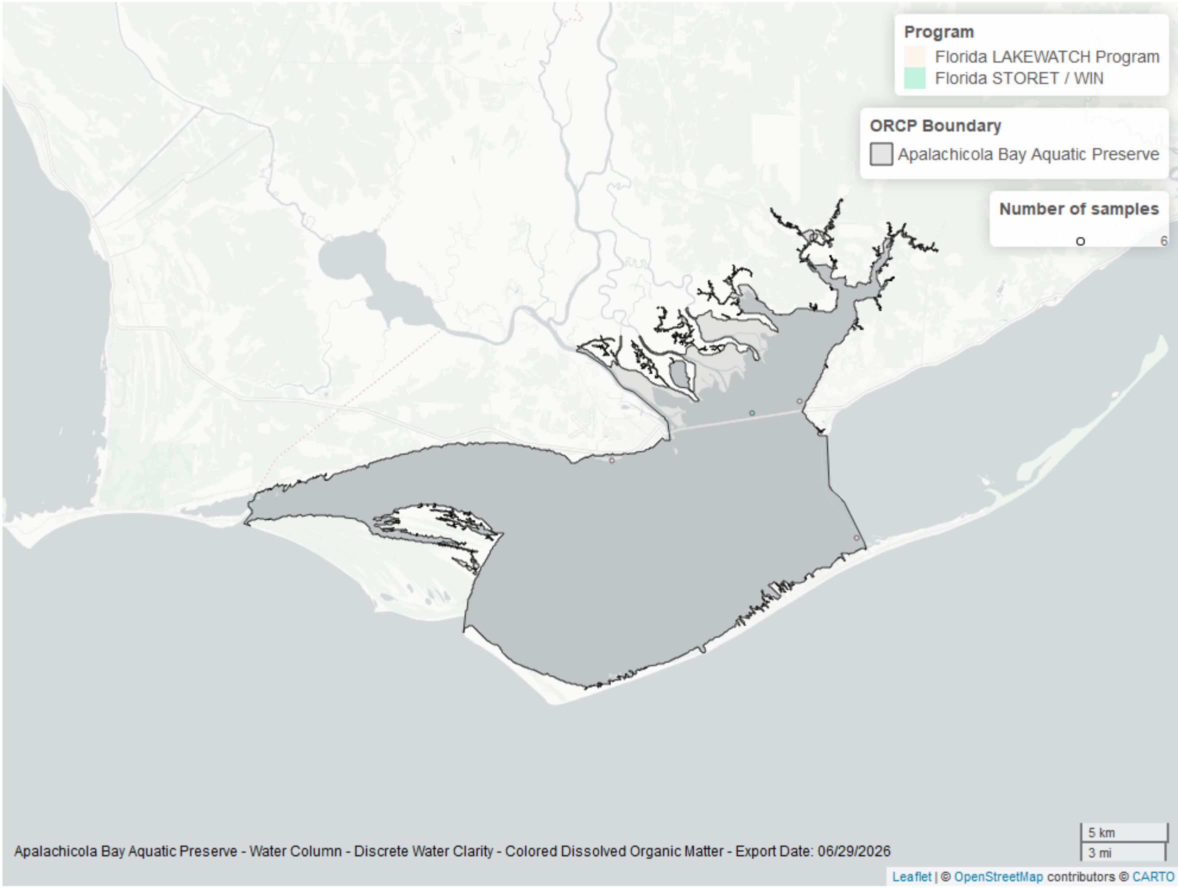

Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.